ITIL®

ITIL® 4: High-velocity IT

Reference and Study Guide

itSMF UK

AXELOS GLOBAL BEST PRACTICE

ITIL® OFFICIAL PUBLISHER

ITIL®

More key titles to enhance your learning

Explore the full range of ITIL® 4 titles



Available in print, PDF and online subscription

- Core textbook per module
- Supporting reference and study guides
- Convenient packs to suit your study pathway towards ITIL 4 Managing Professional or Strategic Leader

Use code NS20 for 20% off today

*applies to print and PDF titles purchased on TSO Shop. Does not apply to apps.

Shop now

bit.ly/ITIL4nextstep

tso
a Williams Lea company

Published by TSO (The Stationery Office), part of Williams Lea,
and available from:

Online
www.tsoshop.co.uk

Mail, Telephone, Fax & E-mail
TSO
PO Box 29, Norwich, NR3 1GN
Telephone orders/General enquiries: 0333 202 5070
Fax orders: 0333 202 5080
E-mail: customer.services@tso.co.uk
Textphone 0333 202 5077

TSO@Blackwell and other Accredited Agents

Copyright © AXELOS Limited 2021

All rights reserved. No part of this publication may be reproduced in any form or by any means without permission in writing from AXELOS Limited.

Applications to reuse, reproduce or republish material in this publication should be sent to the licensing team at: licensing@AXELOS.com

Registered office address: 65 Gresham Street, London, EC2V 7NQ

AXELOS, the AXELOS logo, the AXELOS swirl logo, AgileSHIFT®, ITIL®, MoP®, M_o_R®, MoV®, MSP®, P3M3®, P3O®, PRINCE2®, PRINCE2 Agile®, and RESILIA® are registered trade marks of AXELOS Limited.

ISBN 9780113317981

Printed in the United Kingdom for The Stationery Office.
Material is FSC certified and produced using ECF pulp, sourced from fully sustainable forests.

Figure 4.2: After Snowden, D. (2011) The Cynefin framework. https://cognitive-edge.com/.
The framework is licensed under the Creative Commons Attribution-Share Alike 3.0 unported licence

(https://creativecommons.org/licenses/by-sa/3.0/) from https://commons.wikimedia.org/wiki/File:Cynefin_as_of_1st_June_2014.png

Contents

Foreword v

Acknowledgements vii

About this guide viii

1 Introduction 1

2 Key terms and concepts 3

2.1 Overview of high-velocity IT 3
2.2 Key terms 10
2.3 High-velocity IT objectives 16
2.4 Key characteristics of high-velocity IT 20
2.5 Service-dominant logic 22

3 High-velocity IT across the ITIL model 24

3.1 Four dimensions of service management 24
3.2 Service value system 27
3.3 Service value chain 31
3.4 Digital product lifecycles 36

4 High-velocity IT culture 38

4.1 Key behaviour patterns 38
4.2 Models and concepts 40

5 High-velocity IT techniques 57

5.1 Techniques for valuable investments 57
5.2 ITIL practices contributing to valuable investments 60
5.3 Techniques for fast development 64
5.4 ITIL practices contributing to fast development 74

5.5 Techniques for resilient operations 85
5.6 ITIL practices contributing to resilient operations 93
5.7 Techniques for co-created value 102
5.8 ITIL practices contributing to co-created value 103
5.9 Techniques for assured conformance 107
5.10 ITIL practices contributing to assured conformance 110

6 Taking the HVIT examination 113
6.1 Purpose of the ITIL 4 HVIT qualification 113
6.2 Examination overview 113
6.3 Question type examples 114
6.4 Examination modalities 116

7 The ITIL 4 certification scheme 118
7.1 ITIL Foundation 118
7.2 ITIL Managing Professional stream 119
7.3 ITIL Strategic Leader stream 122
7.4 ITIL Master 124
7.5 ITIL and the T-shaped individual 124

8 ITIL 4: High-velocity IT syllabus 127

End notes 135

Further information 136

Index 139

Foreword

Knowledge is a funny thing: Aristotle wrote, 'The more you know, the more you realize you don't know.'

Over the years I have developed the habit of stretching myself to learn new things. When I am new to a field I find it very frustrating, as there are a lot of new concepts and terms to take in and learn. Eventually I plateau and I find the same search engine results being repeated. This is why, over time, I have switched my learning strategy: instead, I buy several books about a new domain to save myself time in working out the underlying structure of how all these seemingly random concepts fit together. This drastically improves my ability to ask the right questions and delve deeper into a domain. As Donald Rumsfeld put it, I would then go from an unknown unknown to a known unknown state – the perfect position to keep learning forever!

This is also how I see this publication, as the ideal jumpstart to familiarize yourself with the cornerstone concepts of embedding IT into a business strategy. One of the core tenets is high velocity: we are always impressed by people who can do things at a faster pace than we can. Yet there is no instant pill that we can take that will make us learn everything at once. Rightly, as this study guide points out, velocity has a direction, and you as a reader will need to decide where you're going. You may now picture a straight line going from novice to expert, although the mileage to reach your destination may vary. In the same way that agile is incremental and iterative, it's perfectly acceptable to revisit this guide many times and still discover new things. In fact, that is how it was conceived: as an exploration framework.

A great deal of knowledge is distilled in this guide and you will need to unpack it all by reading even more. The knowledge therein has been won by experience and patience, and it has been nicely structured and packaged by Mark, a passionate and seasoned practitioner. Will it work

for you? Hopefully it will inspire you to improve, try different things, and open up your conceptual view of the world.

It will also bring more tools into your everyday knowledge tool belt: every context is different and applying these tools should be done with care. That is why consultants always say 'it depends'. Of course it depends, although that sounds like a broken sales pitch. The trick is to have enough knowledge and tools available to personalize your approach. But before you can do this, you need to learn. This aspect applies not only to you as an individual but to the entire organization in which you work. This study guide will lead you through concepts to bring structure where it is needed, helping you to understand when to loosen the structure, and balance your decisions based on the wisdom of others.

Now let's get to it!

Patrick Debois
Independent IT consultant

Acknowledgements

Lead author

Mark Smalley	Smalley.IT

Reviewers

Akshay Anand	AXELOS
Alison Cartlidge	Sopra Steria
Phil Hearsum	AXELOS
Mark Lillycrop	*it*SMF UK

Contributing authors to *ITIL®4: High-velocity IT*

Akshay Anand, AXELOS; Dan Ashby, Ada Health; Ariana Bucio, BP Gurus; Ana Yasmeen Chong Rosales, BP Gurus; James Finister, Tata Consultancy Services; Reni Friis, Deloitte; Marco Gianotten, Giarte; Rudolf Greger, Design-Thinking-Tank; Ian Jones, Ipswich City Council; Ola Källgården, Olingo Consulting AB; Allan Kelly, Allan Kelly Associates; Donna Knapp, ITSM Academy; Dagfinn Krog, EnterCard; Jeffrey Liker, Liker Lean Advisors; Krikor Maroukian, Microsoft; Simone Jo Moore, SJM; Pasi Nikkanen, HappySignals; Mike Orzen, Mike Orzen & Associates; Oleg Skrynnik, Cleverics; Dave Snowden, Cognitive Edge; Jon Stevens-Hall, BMC Software

About this guide

It comes as no surprise to read that digital technology is increasingly important and therefore places higher demands on those who provide and consume it. You will also recognize the challenges of working in increasingly complex and unpredictable environments. The high-velocity IT concepts, models, and techniques in ITIL 4 represent what happens in highly digitally enabled organizations.

This guide can be used as a study aid for candidates taking the ITIL 4 Managing Professional High-velocity IT qualification and as a quick reference tool for those who just want a high-level overview of the key points from the AXELOS core guidance *ITIL®4: High-velocity IT*. Although the guide does not include exhaustive examinable content, it is a summary for quick revision with specific references to the core guidance should you wish to read more (found in practical tabular format linked to learning requirements in Table 8.1).

Further information is also available in the practice guides on the AXELOS My ITIL platform (one year's subscription to My ITIL is included with any ITIL exam). In addition, you will find useful non-examinable content, including information on taking the examination, tips, key messages, and definitions highlighted throughout, with key figures to help with your understanding. Non-examinable content is marked with the symbol  in the title of the section.

ITIL®4: High-velocity IT introduces you to the ways in which digital organizations and digital operating models function in high-velocity environments. It addresses the specifics of digital transformation and helps organizations to evolve towards a convergence of business and technology, or to establish a new digital organization.

Note that this title is not intended to replace *ITIL®4: High-velocity IT* nor to be a substitute for a course provider's training materials.

1 Introduction

Digital technology is becoming increasingly important. Its economic, societal, and political impacts are unprecedented. At the same time, it is ever-more challenging for digital practitioners to design, develop, run, and support the systems and services that fulfil this demand. *ITIL®4: High-velocity IT* focuses on digital products and services, including digital customer experiences, covering good organizational practices and mental models, all from a practitioner's perspective.

ITIL®4: High-velocity IT is aimed at IT and service management practitioners who work in organizations that are becoming more digitally enabled. It will help those who are familiar with traditional IT and service management concepts to discuss 'digital' confidently, develop practical competencies, and integrate new concepts, techniques, and technologies into their ways of working.

Technological progress, combined with innovatively applied digital technology, has resulted in systems that are complex and unpredictable. Systems now comprise a multitude of components produced by parties who act with varying degrees of dependability, leading to intrinsic fallibility and flaws. Continual changes to a system and its environment mean that these flaws also continually change. Most are too small to cause significant issues, either because of redundancy and other forms of resilience in the system, or because of human intervention.

In the past, most people believed that change disrupted stability, and stability controlled change; fewer changes meant less risk to stability. In the past decade, a different way of thinking has been adopted. By reducing the size of change, the risk of disruption is reduced. Smaller, more frequent changes improve the organization's capability to change,

thereby lowering the risk of disruption. Recently, the prevailing approach has been to embrace and enable changes.

All of this has changed the environment from 'causal' (if-then-else), predictable models to 'dispositional' (if-then-maybe) ones. Because system behaviour cannot always or easily be predicted, ways of thinking and working have shifted: from prediction and control to insight and understanding; from large, occasional changes to small, frequent ones; from detailed planning in advance to continually experimenting and learning; and from failsafe to safe-to-fail. These are non-trivial shifts.

In addition to this increase in complexity-based ways of thinking and working, breaking down silos, or at least silo-thinking, is a current concern. Homogeneous groups are less resilient and effective than diverse ones. In an organizational context, this means that specialists, teams, departments, and organizations should embrace opportunities to work with a wide range of people.

Fortunately, many things are changing. People are working effectively in small, product- or service-oriented teams. Automation is used more often to support IT processes. The interrelated concepts of immutable servers and infrastructure as code (IaC) are widely adopted. Holistic systems thinking is more prevalent, and scientific thinking is being adopted. Unpredictability and ambiguity are increasingly accepted as necessary. There is a shift from a mentality of delivering value to one of co-creating value. External services are often acquired and integrated for the fast development of digital capabilities, as either an alternative or an addition to the internal Agile approach.

These changes are manifestations of new ways of thinking and working that are becoming mainstream. *ITIL® 4: High-velocity IT* builds on these developments, and contributes to them.

2 Key terms and concepts

2.1 Overview of high-velocity IT

The following sections summarize the key features of high-velocity IT (HVIT). They are dealt with in more detail in subsequent chapters.

2.1.1 Key behaviour patterns

To survive and succeed in a high-velocity environment, organizations and people should adopt the following behaviour patterns:

- accept ambiguity and uncertainty
- trust and be trusted
- continually raise the bar
- help get customers' jobs done
- commit to continual learning.

2.1.2 Models and concepts of HVIT culture

To support the key behaviour patterns, organizations should evolve the way they think and operate with regard to:

- **Purpose** How they define and fulfil their mission and objectives
- **People** How they ensure a productive, safe, stress-free environment for their people
- **Progress** How they enable high performance in constantly changing circumstances.

Table 2.1 summarizes the models and concepts that inform an HVIT organizational culture (see Chapter 4 for more detail).

Table 2.1 The models and concepts of HVIT culture

Purpose	People	Progress
Ethics Design thinking	Reconstructing for service agility Safety culture Stress prevention	Working in complex environments Lean culture ITIL continual improvement model (Toyota Kata, OODA loop)

2.1.3 ITIL guiding principles

HVIT organizations benefit from the application of the ITIL guiding principles to their specific needs and circumstances. The seven principles are:

- Focus on value
- Start where you are
- Progress iteratively with feedback
- Collaborate and promote visibility
- Think and work holistically
- Keep it simple and practical
- Optimize and automate.

2.1.4 HVIT approaches

There are many approaches that can be taken to reach and maintain HVIT. Four characteristics are dominant in common HVIT approaches.

These are:

- Lean
- Agile
- resilient
- continuous.

When combined and used together properly by organizations, these characteristics enable the co-creation of value.

2.1.5 Mission, objectives, and techniques

Table 2.2 provides examples of techniques that help significantly different businesses and operations, strategically enabled by digital technology, to achieve their mission and objectives. The techniques are grouped according to their relationship with one of the five HVIT objectives:

- valuable investments
- fast development
- resilient operations
- co-created value
- assured conformance.

Table 2.2 HVIT objectives and associated techniques

Valuable investments	Fast development	Co-created value	Resilient operations
Prioritization (cost of delay; buy/sell/hold; other techniques) Minimum viable products and services Product or service ownership A/B testing	Infrastructure as code Loosely coupled IS architecture Reviews Retrospectives Blameless post-mortems Continual business analysis CI/CD Continuous testing Kanban	Service experience	Technical debt Chaos engineering Definition of done Version control AIOps ChatOps Site reliability engineering
Assured conformance			
DevOps Audit Defense Toolkit/DevSecOps/Peer review			

2.1.6 Practices and their relevance to the five objectives

Table 2.3 indicates the relevance of the 34 ITIL practices to the five HVIT objectives.

Table 2.3 The ITIL practices and their relevance to the HVIT objectives

	Valuable investments	Fast development	Resilient operations	Co-created value	Assured conformance
Architecture management					
Availability management					
Business analysis					
Capacity and performance management					
Change enablement					
Continual improvement					
Deployment management					
Incident management					
Information security management					
Infrastructure and platform management					
IT asset management					
Knowledge management					

Table continues

Table 2.3 *continued*

	Valuable investments	Fast development	Resilient operations	Co-created value	Assured conformance
Measurement and reporting					
Monitoring and event management					
Organizational change management					
Portfolio management					
Problem management					
Project management					
Relationship management					
Release management					
Risk management					
Service catalogue management					
Service configuration management					

	Valuable investments	Fast development	Resilient operations	Co-created value	Assured conformance
Service continuity management					
Service design					
Service desk					
Service financial management					
Service level management					
Service request management					
Service validation and testing					
Software development and management					
Strategy management					
Supplier management					
Workforce and talent management					

High relevance Medium relevance Little or no relevance

2.2 Key terms

2.2.1 Digital organization



Definition: Digital organization

An organization that is enabled by digital technology to do business significantly differently, or to do significantly different business.

Digital organizations are enabled by digital technology. Digital technology is a significant underpinning enabler for these organizations' internal processes, and is often part of their products and services. As such, digital technology is a strategic part of a digital organization's business model, and is applied to its primary (rather than supporting) activities. Because of this, prioritizing digital technology ('digital first') is often part of such an organization's culture (in other words, the way things are done in that organization).

2.2.2 High-velocity IT

Just as some digital organizations are more digital than others, the velocity in some organizations is higher than in others. However, an organization with a higher velocity is not necessarily better. The velocity at which an organization should operate depends on the nature of that particular organization, and in some cases a lower velocity may be more beneficial. It is also not necessary, or even recommended, that the whole of an organization's IT should be high velocity.



Definition: High-velocity IT

The application of digital technology for significant business enablement, where time to market, time to customer, time to change, and speed in general are crucial. High velocity is not restricted to fast development; it is required throughout the service value chain, from innovation at the start, through development and operations, to the actual realization of value.

High velocity does not come at the expense of the utility or warranty of the solution, and in general, high velocity equates with high performance. Increasing velocity within an organization will always involve costs and risks, particularly when there is a steep change rather than gradual improvement.

Scientifically speaking, velocity also has a directional component. In HVIT, this is interpreted as doing the right thing. In other words, not only should the requirements of the high-velocity approach be fulfilled, but the right decisions should be made regarding investment and sustainability.

2.2.3 Digital transformation



Definition: Digital transformation

The use of digital technology to enable a significant improvement in the realization of an organization's objectives that could not feasibly have been achieved by non-digital means.

'Digital transformation' is often used to indicate major investment in digitizing, robotizing, and other forms of automation that enable organizations to do business significantly differently, or do significantly different business. This technological change often requires organizational change in how the organization uses the digital solutions.

The term 'digital transformation' is not specific to a particular type of transformation, and can be used to refer to any transformation that is digitally enabled. The transformed entity is often a combination of the organization's customer experience, products and/or services, business model, operating model (for example, the value stream), and employee experience (see Figure 2.1).

2.2.4 IT transformation

In organizations where business and IT are regarded as separate organizational functions, 'IT transformation' is often used to denote major change that improves how IT services are provided. IT transformation is focused on how IT services and information systems are developed, run, and supported. This can include decentralizing the IT function and integrating it into digital lines of business.

Before they undergo a digital transformation, organizations are managed separately from their IT service providers, whether internal or external. An IT service provider is focused on the management of IT resources to create and deliver IT products and services, whereas a service consumer is focused on the management of its products, services, and resources, including those delivered or supported by the IT service provider. Service consumer organizations may influence the management of the service provider.

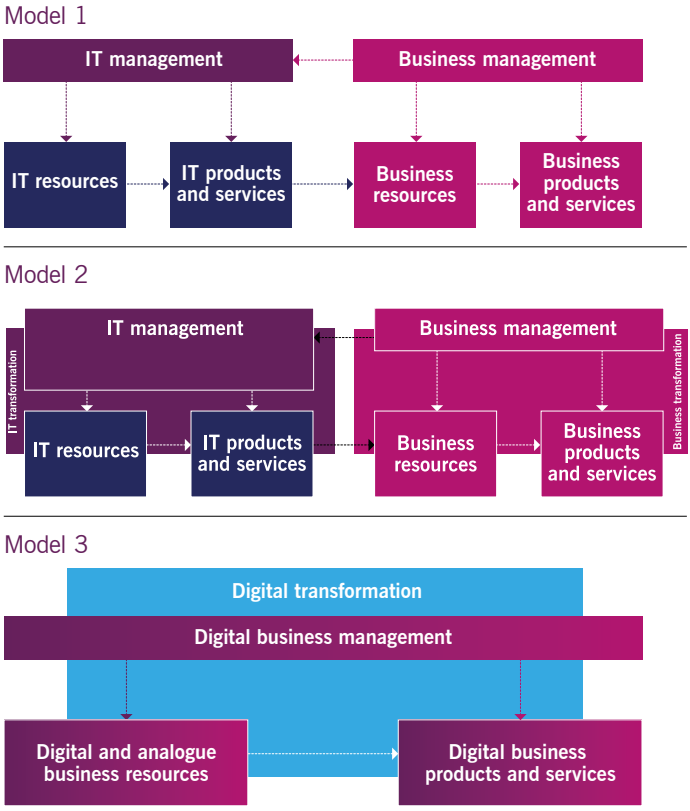


Figure 2.1 Digital transformation and IT transformation

Both the IT service provider and service consumer can transform their management, resources, products, and services. These transformations can be interrelated, but they do not significantly change the way these organizations work together, or the role of IT in the service consumer organization.

Where business and IT are not regarded as separate organizational functions, as is the case in most digital organizations, 'IT transformation' is not an appropriate term to use. 'Digital transformation' would be used instead.

2.2.5 Digital technology



Definitions

- **Digital technology** Technology that digitizes something or processes digital data. Digital technology refers to information technology (IT) and the parts of operational technology (OT) that have been digitized.
- **Digitization** The process of transforming something (e.g. text, sound, or images) from analogue to digital form by expressing the information in binary digits.
- **Information technology** The application of digital technology to store, retrieve, transmit, and manipulate data (data processing), often in the context of a business or other kind of organization.
- **Operational technology** The application of digital technology for detecting or causing changes in physical devices through monitoring and/or control.

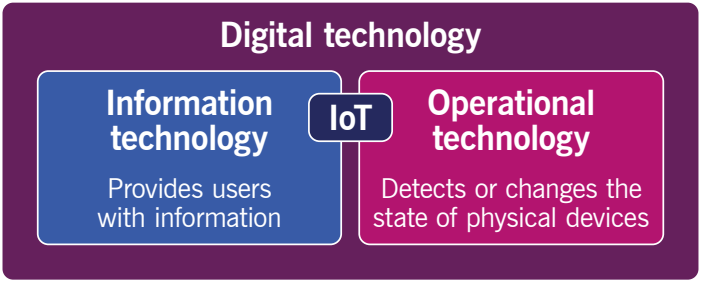


Figure 2.2 Digital technology

Digital technology is made up of both IT and OT. IT provides users with data and information, whereas OT detects or causes changes in physical devices (see Figure 2.2).

2.2.6 Digital product

Definition: Digital product

A product is digital when digital technology plays a significant role in its goods, resources, or associated service interactions.

ITIL defines a product as a configuration of an organization's resources designed to offer value for a consumer. The product is provided to the consumer by means of services that enable the co-creation of value by both parties. These services manifest themselves to the consumer as a combination of acquiring goods, using the provider's (tangible) resources, and interacting with the provider. The value is co-created during service provision and consumption, when the consumer applies

its own resources to use the product. Some products can be defined as digital. In the sense of the definition of digital product, a significant role usually means that without the technology, the product either does not exist or lacks critical features or characteristics.

Services are based on one or more products, and may offer digital and analogue user experiences.

2.3 High-velocity IT objectives

Technology is strategic to digital organizations' business models, and as such, higher demands are placed on the lifecycle of their digital products. These demands can be represented by five high-level objectives that translate the vision and strategy of an organization into more operational objectives and indicators. These objectives are:

- **Valuable investments** The strategically innovative and effective application of IT
- **Fast development** Quick realization and delivery of IT services and IT-related products
- **Resilient operations** Highly resilient IT services and IT-related products
- **Co-created value** Effective interactions between service provider and consumer
- **Assured conformance** To governance, risk, and compliance (GRC) requirements.

It is important to remember that these objectives should not be managed in isolation; they influence and interact with each other, and need to be managed collectively.

In some instances, there will be conflicts between the different objectives. For example, fast development could negatively affect

resilient operations where insufficient time is given to ensuring that services and products are robust. For this reason, it is important to ensure that the objectives are properly balanced.

2.3.1 Valuable investments

The valuable investments objective involves identifying and justifying digital investments that would contribute significantly to business strategy. This exercise should result in a good understanding of a digital investment's potential value, anticipated costs, and return on investment, and the defined criteria for its utility. The warranty, or non-functional requirements, should also be determined, although it does not add to the potential value of the functionality. Warranty ensures that an investment's potential value is not adversely affected by outages, poor use, or other factors.

Making valuable investments is founded in market research and the development of new products. New digital products and services should be envisaged and evaluated in terms of profitability.

The value of an investment can only be determined when the return is realized; this occurs after the investment, when value is co-created between the provider, the consumer, and other stakeholders.

In digital organizations, IT drives and enables the business. It is therefore of crucial importance that the ever-evolving potential of IT is continually evaluated for strategic advantage.

2.3.2 Fast development

The fast development objective involves realizing new and improved digital products and services frequently, quickly, and reliably. 'Development' refers to product development in general, although application development is often included in this.

In general, the sooner digital products are delivered, the sooner value can be realized. Sometimes, however, this is not the case, and the schedule should be amended accordingly; for example, an early delivery might not align with market demand. Separating a single product into a series of incremental deliveries enables faster overall delivery, and allows users to realize value earlier than if they wait for the whole product.

Under business pressure to realize value quickly, Agile development teams deliver potentially deployable software increments as frequently as possible. Unfortunately, these releases often have to wait for days, weeks, or sometimes months for actual deployment, which can be the longest delay in the value stream. This usually occurs because of a widespread cautious approach to approval and deployment. Therefore, there is a potential conflict between fast development and resilient operations.

2.3.3 Resilient operations

The resilient operations objective involves ensuring that digital products are available for use whenever needed.

The potential value of digital investments can only be realized when the digital products and services that have been invested in are available for use. Fulfilment of the non-functional requirements provides warranty, and reduces the risk that issues will adversely affect the utility of the products and services.

Information systems increasingly rely on so many components that behaviour often cannot be predicted or guaranteed. Failsafe systems are an illusion; organizations must be prepared for inevitable and unexpected failure. The emphasis is no longer on maintaining a long interval between failures; it is on restoring service quickly when inevitable issues do occur. This reduces the disruption to business operations.

2.3.4 Co-created value

The co-created value objective involves co-creating value from digital products through the close collaboration of the service provider and the service consumer.

Co-created value is about the service consumer using the service provider's products and services effectively and benefiting from their utility and warranty.

In HVIT environments, where service consumers use digital products and services regularly, expectations are higher. Consumers demand a more intuitive and responsive user and customer experience (UX and CX). The better the service providers understand how service consumers use the IT service or digital product, the better equipped they are to support them. Equally, the more that service consumers understand how service providers provide IT services or digital products, the better equipped they are to interact effectively with them.

2.3.5 Assured conformance

The assured conformance objective involves ensuring that service provision and service consumption comply with corporate and regulatory directives with respect to governance, risk, and compliance. Beyond ensuring conformance, it is also important to assure the accountable people that conformance has been achieved.

Although external requirements may remain the same, there may be alternative and more appropriate ways for digitally enabled organizations to fulfil them.

High velocity is often associated with taking risks and, from a commercial perspective, these risks may be necessary. Nevertheless, risks must be justified, and there are internal rules and external regulations that organizations must comply with. Assured conformance

reassures those people who are accountable for, and affected by, governance, risk, and compliance issues, as they feel more confident knowing that the organization acts within these constraints.

2.4 Key characteristics of high-velocity IT

There are many approaches that can be taken to reach and maintain HVIT. Four characteristics are dominant in common HVIT approaches. These are:

- Lean
- Agile
- resilient
- continuous.

When combined and used together properly by organizations, these characteristics enable the co-creation of value. The benefits of each characteristic are outlined in Table 2.4.

When used together, these characteristics lead to the co-creation of value, extending the focus of an approach to effective service consumption. Value is only realized when the user actually uses the digital products and services. An approach that uses all four key characteristics of HVIT will help the service provider to ensure that the service consumer achieves the desired outcomes.

2.4.1 Lean

Approaches with Lean characteristics focus on breaking down large pieces of work into smaller batches. A short lead time is the best way to ensure quality, customer satisfaction, and employee happiness, and can be achieved by using small batch sizes of work. It is therefore beneficial to break down larger pieces of work into smaller ones. This will often present a challenge to work that is organized into larger batches which are passed between functions with formal handover procedures.

Table 2.4 Key characteristics of high-velocity IT

Characteristic	Benefits
Lean	Helps to improve throughput and reduce waste. HVIT environments benefit from approaches with Lean characteristics due to the pressure on time to market and time to customer.
Agile	Adds close and iterative collaboration with users. Approaches with Agile characteristics are important for HVIT environments because digital products and services have to be developed in response to changeable market demands.
Resilient	Maintains workable availability and performance. The systems that support HVIT environments are complex and therefore error-prone. Approaches with resilient characteristics minimize the effect of incidents by degrading systems gradually and restoring service quickly.
Continuous	Ensures fast and reliable deployment. Approaches with continuous characteristics extend the Lean focus on throughput by standardizing and automating processes for integrating, building, testing, and shipping code, enabling digital products and services to be available when required.

2.4.2 Agile

Building on the Lean principle that small batch sizes of work are beneficial to throughput, approaches with Agile characteristics focus on delivering small iterations of products or services, in frequent increments, so that the approach can be adjusted in response to changes in the environment. In these approaches, feedback, in the form of information, is gathered as quickly as possible and decisions are delayed as long as possible.

Agile techniques are focused on ongoing conversations and interactions between software developers, business people, and other stakeholders who improve the customer experience.

2.4.3 Resilient

Approaches with resilient characteristics are focused on maintaining workable availability and performance, and minimizing the effect of incidents. Two examples of approaches with resilient characteristics are site reliability engineering (SRE) and DevOps.

SRE applies a software development mindset to IT operations, and helps to bridge the gap between development and operations. SRE teams are created alongside existing teams for IT operations.

DevOps and, more explicitly, DevSecOps, promote the integration of security into the daily work of application development and IT operations, rather than having it as a separate 'policing' function.

2.4.4 Continuous

Approaches with continuous characteristics, such as continuous integration, delivery, and deployment (CI/CD), are based on the belief that small and frequent batches of work are not only more valuable because functionality can be used earlier, but are safer because the change is smaller and feedback is obtained sooner.

2.5 Service-dominant logic



Definition: Service-dominant logic

A mental model of an (economic) exchange in which organizations co-create value by applying their competencies and other resources for the benefit of each other.

Service science defines service as the application of resources (including competencies, skills, and knowledge) to make changes that have value for another organization. A core concept in service science is service-dominant logic.

Service-dominant logic regards service as the process of doing something for and with another party. Value creation is a collaborative process. In service-dominant logic, value is always co-created. Service-dominant logic contrasts with goods-dominant logic, in which the provider delivers value to the customer by transferring ownership of goods.

2.5.1 Application

When service-dominant logic is applied to service delivery, the provider pays more attention to the customer's specific situation, and involves the customer in the service delivery process. This is a more effective way of helping to get the customer's job done. The service interaction is consumer-focused, and the consumer creates value when they integrate and apply the resources of the service provider (and of other service systems) to achieve the exchange. Each consumer determines the value or quality of the service experience based on personal needs at a specific time and in a particular context.

3 High-velocity IT across the ITIL model

3.1 Four dimensions of service management

The four dimensions of service management identify the kinds of resources that are used in a value stream. Three of the four dimensions (organizations and people, information and technology, and partners and suppliers) are concrete resources that are used operationally during the execution of the value stream steps. The value streams and processes dimension represents abstract resources that are used as input for the design of value streams. This is shown in Figure 3.1.

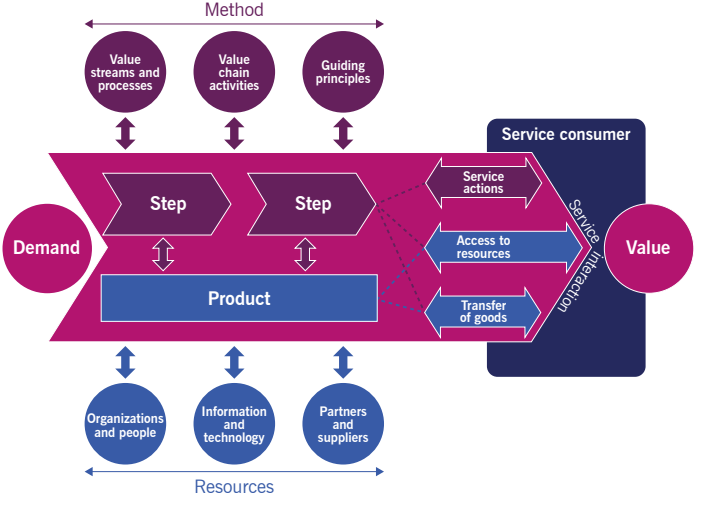


Figure 3.1 The value stream in context

Examples of the application of each dimension to HVIT are described in the following sections.

3.1.1 Organizations and people

In HVIT environments, where IT is an integral part of an organization's products and services, it is likely that the IT function will be an integral part of the lines of business that are responsible for the various products and services. There will often be multifunctional product/service-based teams, with both business-oriented and IT-oriented team members. Although there may be a centralized IT service centre for non-differentiating digital technology such as email and Wi-Fi, the differentiating digital technology will often be managed as part of the primary activities. There may also be platform-oriented teams that support various decentralized product/service-based teams.

3.1.2 Information and technology

In HVIT environments that derive significant value from digital technology, part of this value is related to the information that the digital systems provide, and part is related to the digital technology that provides the information quickly, resiliently, securely, and efficiently. Higher demands are therefore placed on both the information and technology that digitally enable organizations, and the information and technology (IT tooling) that support the IT processes. An example of HVIT tooling is an automated deployment pipeline that ships new versions of applications faster and more reliably to production.

3.1.3 Partners and suppliers

HVIT environments typically make extensive use of cloud-based infrastructures, platforms, and other services. These utility services are often provided by organizations on their own terms and conditions. Public cloud-based services are usually high quality and affordable, but

the individual consumer has little or no influence on the provider. It is therefore crucial to analyse the dependencies and take appropriate action in terms of contracts and service level agreements (SLAs), including risk-sharing agreements, secondary providers and contingency plans, and workarounds.

When outsourcing work in an HVIT environment, it is important to consider whether the external service provider works in a similar way to its customer, as it is difficult to integrate parties with fundamentally different ways of working into the same value stream. Functional outsourcing, in which discrete functions, such as testing, are outsourced, is often less effective than outsourcing a whole value stream.

3.1.4 Value streams and processes

HVIT environments recognize that it may be better to create a unique value stream for each digital product or service. This may be less efficient than a standardized and centralized single value stream that serves multiple products and services, but the benefits in terms of effectiveness will often outweigh the costs, so it is advisable to consider this alternative.

HVIT environments often deal with complex systems that are unpredictable, so the use of processes should be reserved for those situations where it is feasible to predetermine the appropriate sequence of activities. Practitioners will often think in terms of various patterns of activities that might work, and will experiment to select the right patterns for the task at hand.

3.1.5 External (PESTLE) factors

An organization's choices of markets, products and services, and resources and activities are influenced by multiple external factors, which can be political, economic, social, technological, legal, and environmental (PESTLE). These all influence the four dimensions of service management.

HVIT environments are also characterized by relatively high degrees of volatility, uncertainty, complexity, and ambiguity, which are referred to by the acronym 'VUCA'. VUCA presents a serious managerial challenge, and should be considered in the planning and management of organizations.

An organization that experiences VUCA to a greater degree will typically pursue more evolutionary approaches, based on an assessment of the disposition of the actors in the external and internal environment, and experimentation with the factors that influence those actors. In these organizations, the focus is more on managing the present, rather than making and following a roadmap to a predetermined future state.

3.2 Service value system



Exam tip

It should be noted that there is some overlap between assessment criteria 2.1(b) and 2.1(c): both refer to the service value chain (section 3.3), the service consumer (section 3.3.3), and value streams (section 3.3.1). The content has not been duplicated for these topics, so the reader should read section 3.3 with both assessment criteria in mind (see Table 8.1 for the assessment criteria in the HVIT exam).

It should also be noted that the service value system comprises topics that are described in other sections of this publication. They are the ITIL guiding principles (section 4.2.4 non-examinable), the ITIL practices (sections 5.2, 5.4, 5.6, 5.8, and 5.10), the service value chain (section 3.3), and continual improvement (section 4.2.3.3 (OODA loop, non-examinable)).

HVIT aligns high-value IT-related work with high-velocity business, from innovation to value realization. This requires fast flow, fast feedback, and fast improvement, and not only results in things getting done faster but also improves the quality of IT-related products and services. This has significant implications for organizational IT operating models. Digitally enabled organizations define and structure their IT-related activities and resources differently from organizations in which IT is of less strategic significance. The higher demands that digitally enabled organizations place on technology are reflected in the way they operate.

This section illustrates how ITIL guidance can be used to provide building blocks for an IT operating model that defines HVIT work and its organization.

3.2.1 Operating models

An operating model is the 'back end' of an organization's business model, and delivers the value propositions defined in that business model. The operating model is a representation of the building blocks of an organization and the relationships between them. It can be used as a description of the current state of the organization, or as a future state design, referred to as a 'target operating model'.



Definitions

- **Operating model** A conceptual and/or visual representation of how an organization co-creates value with its customers and other stakeholders, as well as how the organization runs itself.
- **High-velocity IT operating model** An IT operating model where digital technology plays a major role in the co-creation of value.

A high-velocity approach is typically required for an operating model when there is co-creation of value with customers, when operating models are becoming significantly more digital, or in response to uncertain, ambiguous, and rapidly changing circumstances. There is often as much value, if not more, in the process of creating the operating model as in the end product.

An HVIT operating model is one in which digital technology plays a major role in the co-creation of value. All operating models have some digital elements, but a digital operating model is one where digital technology makes possible a model that is otherwise not feasible or practical without it.

Operating models for digitally enabled organizations are characterized by their focus on:

- dedicated value streams for each of their products and services
- a co-creational culture that fosters high performance and continual improvement
- permanent product/service-based teams over temporary project teams
- automation of IT processes, including infrastructure as code (IaC).

In ITIL, value streams are at the core of the operating model because they are the sets of steps required to deliver products and services. HVIT can be supported by many elements of the ITIL service value system (see Figure 3.2).

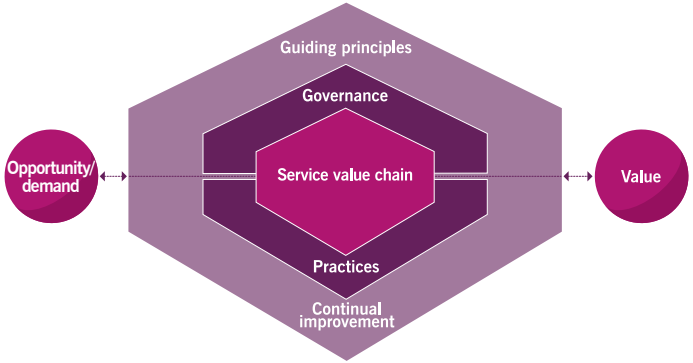


Figure 3.2 The ITIL service value system

3.2.2 Governance and management

Governance is the means by which an organization is directed and controlled. The governing body evaluates the organization's situation, sets the direction for managers, and monitors the organization's performance. Management is anchored in governance. Managers deal with planning, building, organizing, and improving the organizational entity.

Responsibility for digital technology is an integral part of the lines of digital business in digitally enabled organizations, and is not a separate unit such as a centralized IT service centre (although these may exist for non-differentiating IT services, such as email). The governed and managed organizational entity is therefore responsible for both digital technology and its use in the context of digital products and services.

Practitioners operate within a governance and management framework. They understand the applicable constraints and know how to act within that framework, and their insight and judgement influence how they act.

The more insight they have, and the better their judgement skills, the more the practitioner will take the initiative to bend rules when the associated benefits and risks are justifiable. This can be very beneficial, as HVIT environments are often highly unpredictable, meaning that predetermined instructions are neither feasible nor desirable.

An HVIT practitioner therefore has to exercise judgement on the job. To do this effectively, they must understand the reasoning behind certain constraints that are in place. A major role of the manager in an HVIT environment is therefore to provide context and to enable the practitioner to take charge.

3.3 Service value chain

The ITIL service value chain is useful for describing, at a fairly high level of abstraction, the types of activities that a service provider executes (see Figure 3.3). It helps people to focus on the goal of each value chain

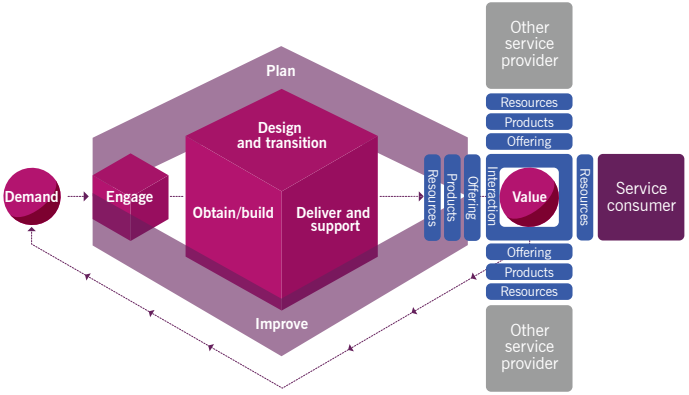


Figure 3.3 The ITIL service value chain

activity, and the inputs and outputs, without getting lost in the details of the lower-level activities in a value stream.

The value chain activities are closely interrelated, and can be arranged in any order to explain and discuss a range of different situations.

3.3.1 Value streams

A value stream is a series of steps an organization undertakes to create products and services and deliver them to consumers. They are sets of activities that are performed according to guidelines (principles, approaches, techniques, etc.) and within constraints (regulations, policies, budgets, etc.).

This is depicted in Figure 3.1, in a value stream that is triggered by demand for value, which is co-created using products and services. The value stream consists of steps in which resources are combined, resulting in products designed to enable value creation, not only for the service consumer but also for other stakeholders. Activities performed by the resources are guided by various methods adopted by the organization (descriptions of value streams and processes/procedures/work instructions in the practices, descriptions of the activities in the service value chain, and the guiding principles).

The activities that are required to deliver products and services are modelled by the organization's service value chain and practices. The service value chain describes the organization's archetypal activities (see Figure 3.3).

The service value chain describes the activities that are required to effectively manage products and services, whereas a value stream comprises an actual series of steps to create products and services and deliver them to consumers. As such, value streams can be considered to be where things actually happen: where the ITIL practices are used and value is co-created. There is no set structure for value streams, and

they are unique to each organization. HVIT organizations are often product/service-oriented and have multiple value streams that reflect the diversity of their products and services. Their operating models therefore comprise multiple value streams.

It is also important to get feedback as quickly as possible, not only to make any improvements that may be needed but, crucially in a volatile HVIT environment, also to evaluate whether the way of working or way of thinking needs to be improved. In these dynamic environments, the improvement of daily work is as important as actually doing daily work.

In HVIT environments, systems are often complex and therefore unpredictable. This makes it less likely that detailed processes, procedures, and work instructions will be useful, as often they will not be followed. It is also not useful or feasible to predict or dictate the sequence of steps in a value stream, and the activities within those steps, other than at a high level of abstraction. Instead, the sequence of activities and steps will often emerge during, and as a result of, the 'micro-interactions' that take place during execution. This means practitioners must be able to observe the expected and unexpected changes that their actions, and those of others, make, and adjust their next actions accordingly.

3.3.2 Service value chain activities and DevOps

In HVIT environments, the concept of 'continuous' is often used, and is a key characteristic of many HVIT approaches. 'Continuous' refers to fast, iterative cycles of activities, enabled by short feedback loops. CI/CD (continuous integration, continuous delivery, and continuous deployment) is a well-known example of this, as used by the DevOps community.

Figure 3.4 shows how the DevOps activities (in the loop) and the service value chain activities can be combined. Some service value chain activities (such as design and transition) have been split into two 'sub-activities' that are easier to map to parts of the DevOps model.

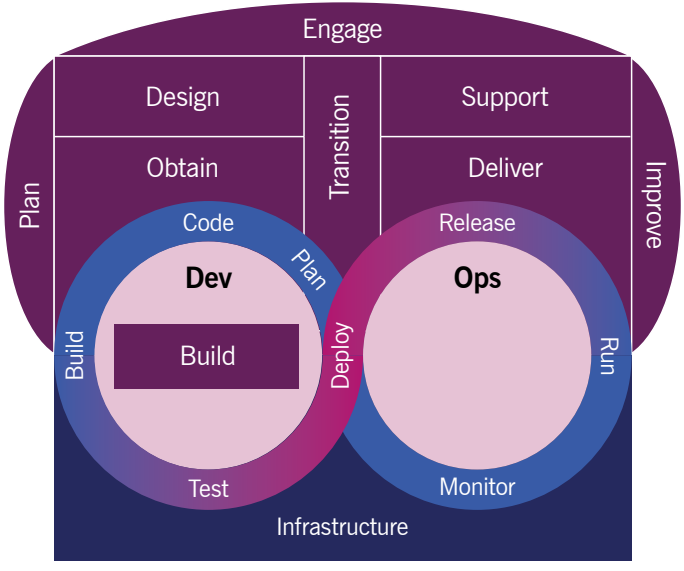


Figure 3.4 DevOps and the service value chain

In ITIL terminology, DevOps focuses on developing, deploying, and running concrete service components rather than intangible services. The dominant service components in DevOps are the applications, data, and platforms that together form a product for the consumer. The focus of the service value chain is on products and services rather than individual service components. It describes what is needed for the interaction between the service provider's products and other resources, including the service consumer's resources.

3.3.3 Service consumer

Another way that HVIT can be applied to the service value chain is in illustrating how closely service providers and service consumers interact. A service consumer regards the high-level interaction with a service provider in terms of acquisition, provision, and use. The consumer translates their demand into requirements, and engages a provider to provide services that they then use to create value.

Figure 3.5 shows the six value chain activities of the service provider, with initial design and transition, and obtain/build in the top half, and deliver and support (in parallel with subsequent enhancement) in the bottom part. The primary interfaces between the consumer and provider are marked by the consumer's requirements (R) and the provider's service (S). Design and development, and deliver and support are connected by the product (P) that is deployed.

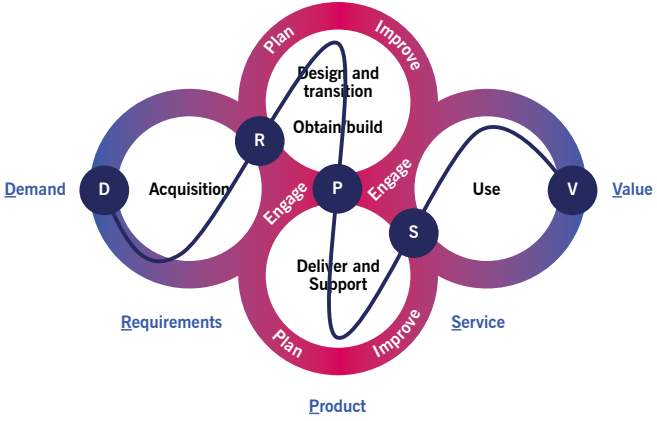


Figure 3.5 An example of a value stream referring to service value chain activities

Figure 3.5 also shows how a value stream can flow across the value chain activities. It is important to recognize that the service value chain and associated practices are models on which actual value streams are based. Value streams can be observed where people and other resources act to co-create value. Service value chains and associated practices cannot be observed, as they are abstract representations that are used to model the value streams.

3.4 Digital product lifecycles

Service consumers and service providers have different perspectives on digital products. They each have their own product lifecycles, which overlap during the period of engagement between the consumer and provider. For the service provider, the lifecycle of a product lasts for as long as there are potential customers for that product. For the service consumer, the lifecycle continues for as long as the product is used, and strictly speaking is a product use lifecycle.

Table 3.1 outlines the lifecycle stages of a digital product from the perspective of the service provider and service consumer.

Table 3.1 Stages of the digital product lifecycle

Lifecycle stage	Service provider	Service consumer
Exploration	The service provider researches and develops the product and service offering.	The service consumer becomes aware of the existence of a product and assesses it as interesting and then desirable, after which an agreement is made.
Onboarding	An instance of the product is installed, and the user organization is onboarded, sometimes with a transition from a replaced product.	
Co-creating value	The service provider delivers and supports the product and experiences increasing, stable, or decreasing return on investment, leading to an investment decision to buy (use and improve the product), hold (use but do not improve it), or sell (use and reduce, replace, or retire it).	The service consumer uses the product and experiences increasing, stable, or decreasing value, eventually leading to a decision to replace or retire the product.
Offboarding	The instance of the product is uninstalled and the user organization is offboarded, sometimes with a transition to a replacement product.	
Retired	The service provider no longer offers or supports the product.	The service consumer no longer uses this product, but other consumers may use it.

4 High-velocity IT culture

This chapter explores the kind of culture that supports and enables HVIT work. It outlines the five key behaviour patterns that reflect the organizational needs and the aspirations of practitioners to work in a rewarding environment. It also explores models and concepts that support these behaviour patterns and inform organizational culture. The supportive relationship between models and concepts and behaviour patterns is not binary: most models and concepts support various behaviour patterns in varying degrees, and the more significant relationships are indicated in Table 4.1.

4.1 Key behaviour patterns

There are five key behaviour patterns that reflect organizational needs and the aspirations of practitioners to work in a rewarding environment. These behaviour patterns are intended to appeal to humans' desire to contribute to something worthwhile, to learn and improve, and to be recognized for their intent and effort.

The five key behaviour patterns are shown in Figure 4.1.

Although most of these behaviour patterns are not exclusive to HVIT, the combination of and adherence to all five is beneficial to the kind of people who understand the demands of a more digitally enabled enterprise.

4.1.1 Accept ambiguity and uncertainty

This behaviour pattern reflects the volatile and ambiguous nature of typical digitally enabled organizations. In these environments, where failsafe systems are often an illusion, it is important to favour

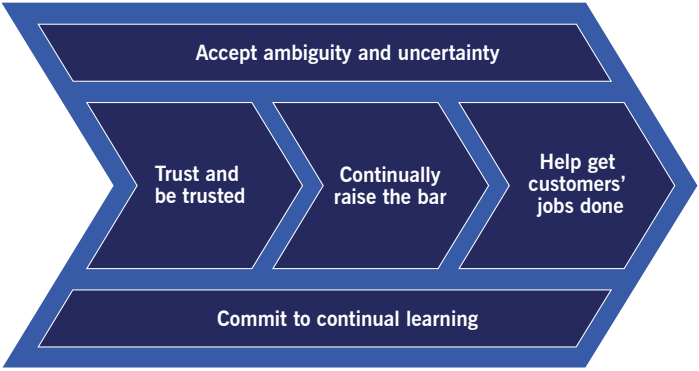


Figure 4.1 Key behaviour patterns

experimentation and not be afraid to fail safely and mindfully. Practitioners should not be scared of the unknown.

4.1.2 Trust and be trusted

This behaviour pattern is about respecting people's professional skills and trusting them to take the right decisions, along with being considerate to others, acknowledging diversity, and being inclusive. People should be given feedback on their work that is honest, but considerate.

4.1.3 Continually raise the bar

This behaviour pattern focuses on never being satisfied with the status quo, with the viewpoint that even if things are satisfactory today, they will not be tomorrow.

4.1.4 Help get customers' jobs done

This behaviour pattern represents the essence of every organization. All organizations have a range of stakeholders, but customers are typically the most important. This behaviour pattern focuses on the act of helping somebody solve their problems and become who they would like to be.

4.1.5 Commit to continual learning

This behaviour pattern underpins the others. It is important that practitioners commit themselves to continually learning and improving their knowledge and the level of information they have. Data-driven experiments can be used to challenge and improve hypotheses.

4.2 Models and concepts

There are a number of models and concepts that inform organizational culture. These are grouped into three categories:

- purpose
- people
- progress.

Table 4.1 outlines the key behaviour patterns to which each of the models and concepts, and each of the ITIL guiding principles, relates.

Table 4.1 Models and concepts and related key behaviour patterns

	Accept ambiguity and uncertainty	Trust and be trusted	Continually raise the bar	Help get customers' jobs done	Commit to continual learning
Purpose					
Ethics	✔	✔	✔	✔	✔
Design thinking				✔	
People					
Reconstructing for service agility			✔		
Safety culture		✔			
Stress prevention		✔			
Progress					
Working in complex environments	✔				
Lean culture		✔	✔		✔
ITIL continual improvement model	✔		✔		✔
ITIL guiding principles					
Focus on value				✔	
Start where you are	✔				

Table continues

Table 4.1 *continued*

	Accept ambiguity and uncertainty	Trust and be trusted	Continually raise the bar	Help get customers' jobs done	Commit to continual learning
Progress iteratively with feedback	✔				✔
Collaborate and promote visibility		✔			
Think and work holistically	✔				
Keep it simple and practical			✔		
Optimize and automate			✔		

4.2.1 Purpose

Purpose is the external goal that drives people's efforts: that is, the changes that products and services enable.

4.2.1.1 Ethics



Definition: Ethics

A system of principles that defines what is good for individuals and society.

The societal, political, and economic impact of IT is unprecedented, for better and for worse. Digital organizations, where IT drives the business rather than just supports it, therefore have an increasingly strong moral obligation to consider how they apply IT, beyond their direct economic interests.

Software engineering often becomes a means of social engineering, and this is not always intentional. Machine learning algorithms are susceptible to taking a training data set and amplifying the worst of human biases. Organizations have a responsibility to behave ethically in order to understand and correct such biases.

Application

When applying ethics, practitioners should aspire to the following behaviours:

- Think about how their actions affect others.
- Establish generic ethical principles.
- Accept that ethical principles simply help to clarify specific situations.
- Discuss dilemmas.
- Take responsibility for choosing the least worst course of action.

4.2.1.2 Design thinking



Definition: Design thinking

The set of cognitive and practical processes by which design concepts are developed.

For digitally enabled organizations, the quality of their digital products and customer experiences is immensely important. An understanding of design thinking gives practitioners, together with co-workers and customers, the ability to contribute to the creation of better digital products and customer experiences, and help to complete jobs for customers effectively.

Application

When applying design thinking, practitioners should aspire to the following behaviour:

- **Empathize with stakeholders** It is important to be able to understand the point of view and needs of stakeholders.
- **Speculate and experiment** Designers should be able to create hypotheses based on observation and reflection, and test them with prototypes.
- **Dare to decide** Designers should focus on what customers do and need, not necessarily on what they say they want. Designers should be empathetic and able to feel what is most important, and make decisions accordingly.

4.2.2 People

The models and concepts in the people category are about organizing work that produces or relies on knowledge, and fostering a healthy and productive workplace.

4.2.2.1 Reconstructing for service agility



Definition: Reconstructing for service agility

An approach to organizing knowledge work and service provision that reflects its complex and social nature.

HVIT ways of working have serious implications for an organization and the way it manages its services. Management is traditionally focused on specialization of work, prescriptive processes, and targets to drive and measure performance, based on the assumption that resources and work are predictable enough to be controlled at a low level of granularity.

Within knowledge work and service provision, however, there are many subtle human interactions that make it impossible to predict the outcomes of a service interaction. This makes it pointless to impose detailed targets on humans as if they were machines. People are intrinsically unpredictable, and will respond to the perceived intent of a co-worker or customer, who will then respond in turn. Sometimes the intent is incorrectly perceived, leading to an unexpected response. Both parties must therefore constantly assess their responses and adjust accordingly.

Where humans are involved, service interactions are social interactions. A typical management approach does not reflect the social reality of a service workplace, and is therefore ineffective. A better approach[1] is to loosen control so that people have more freedom to use their professional judgement in intrinsically unpredictable circumstances. This is central to the idea of reconstructing the organization for service agility.

Application

When reconstructing for service agility, practitioners should aspire to the following behaviour:

- **Loosen control** Give people more freedom to use their professional judgement in intrinsically unpredictable circumstances.
- **Use process as temporary scaffolding to provide support** Processes and other forms of control can provide support until there is enough competence and confidence to work heuristically with fewer constraints.
- **Respect team members' values and allow them to work within these** People make their best contributions to work that is aligned with their personal values.

4.2.2.2 Safety culture



Definition: Safety culture

A climate in which people are comfortable being (and expressing) themselves.[2]

In a safety culture, people feel trusted and valued. They are therefore more likely to point out risks than when they fear that this would damage their reputation and position. In HVIT environments, it is crucial that people feel able to share their opinions and experiment with improvement without the fear of judgement or embarrassment.

Application

When applying safety culture, practitioners should aspire to the following behaviour:

- **Act on safety requirements** Don't just talk about why safety is important: do something about it.
- **Exhibit vulnerability** Don't be afraid to speak up about any doubts, and ask for help when needed.
- **Foster feedback and act on it** Encourage people to share their opinions.
- **Be kind and compassionate** Build human relationships.
- **Be realistic about failure** Acknowledge that failure will happen and that people are not to blame, but the system.

4.2.2.3 Stress prevention



Definition: Stress prevention

The prevention, monitoring, and remediation of unhealthy tension in the workplace.

Many people suffer from stress, burnout, and mental health issues. Although physical health issues have been recognized and talked about at work for a long time, discussion of mental health issues has only seriously begun more recently. It is now, however, becoming acceptable as well as desirable to address such issues. Mental health concerns are not exclusive to HVIT or IT in general. They are, however, critical for both high organizational performance and the well-being of employees.

As delivery models adopt high-velocity approaches, organizations need to change the way that they create and maintain products and services. In some cases, this may require new technologies, but in all cases, there will be a significant shift in culture, performance measurement, and other human elements; hence it becomes even more important to consider the human impact. This applies both during the transformation of the model and once the model has become the norm. These two stages present very different but equally serious challenges that many organizations are yet to address.

Application

When looking to prevent stress in the workplace, practitioners should aspire to the following behaviour:

- Elicit advice from mental health professionals.
- Conduct a risk assessment of mental health issues.
- Develop strategic, tactical, operational, and non-judgemental approaches to deal with workplace stress and mental health.
- Ensure that the approaches adopted are only used for the welfare of individuals, not subverted for other objectives.
- Help those with invisible disabilities to interface with HVIT teams effectively.

4.2.3 Progress

The models and concepts in the progress category are about understanding the complex nature of work, and improving by learning.

4.2.3.1 Working in complex environments



Definitions

- **Systems thinking** A holistic approach to analysis and decision-making that focuses on the relationship between a system's components and the way the system works, both as a whole and within the context of larger systems.
- **Complexity thinking** A systems thinking approach based on the recognition and understanding of the various levels of complexity inherent in the systems and the context in which they operate.
- **Complex adaptive systems** Systems that adapt in, and co-evolve with, a changing environment, resulting in:
  - behaviour that is not predicted by the behaviour of parts of the system
  - the inability to examine the system in isolation from the other systems in its environment.

In order to work effectively, practitioners must understand the nature of the environment, or system, in which they work. A major characteristic of a system is its predictability, as this determines which way of working is effective. Systems thinking in general, and complexity thinking in particular, help make sense of the system and offer guidance as to effective approaches.

Application

When working in complex environments, practitioners should aspire to the following behaviour:

- Assess the causality, while being aware of natural bias towards the domain of dominant experience.
- Act according to the context; in terms of the sense-making Cynefin framework[3] (Figure 4.2):
  - In the Clear* domain, apply constructs such as prescriptive processes and detailed waterfall-based plans.
  - In the Complicated domain, apply constructs such as case management[4] and timeboxing. When transitioning from Complex to Complicated, apply constructs such as Scrum.
  - In the Complex domain, apply 'pre-Scrum' techniques such as parallel/independent experimentation and disintermediation of the analyst to discover and resolve unarticulated or ambiguous user requirements.
  - In the Chaotic domain, take control and act quickly to stabilize the situation and transition it into the Complex domain.
- Beware of using 'recipes' based on cases that are presented as if they can be adopted without adaptation; understand the context and do not confuse correlation with causation (sometimes things do not repeat themselves in the same context); never repeat 'what you do' without understanding 'why you do it'; do not blindly copy industry leaders.
- Be careful when defining the future state and closing the gap; do not try to engineer human systems, but evolve them; develop the evolutionary potential of the present and move to the 'adjacent possible'.

* The Clear domain was previously known as Obvious, and before that, as Simple; however, there is no change to the substance of the domain.

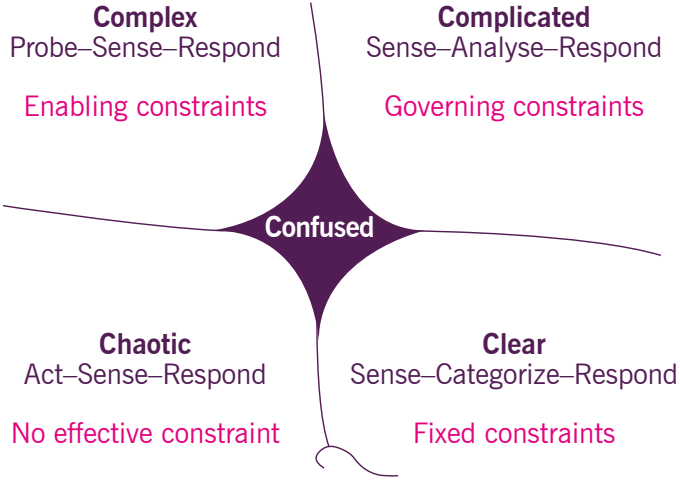


Figure 4.2 The Cynefin framework
After Snowden (2011); reproduced by permission of Cognitive Edge

4.2.3.2 Lean culture



Definition: Lean culture

A work environment where trust, respect, curiosity, enquiry, playfulness, and intensity all co-exist to support learning and discovery.

Lean is a balance between striving for standardization and predictability to avoid errors, and at the same time fostering a culture of calculated risk-taking, curiosity, and enquiry, based on a foundation of trust and respect.

Application

When applying Lean culture, practitioners should aspire to the following behaviour:

- Trust people and the 'system', but remain vigilant and give feedback when needed.
- Treat people decently.
- Strive to understand how things work and what could be improved.
- Gather facts systematically and challenge hypotheses.
- Develop new insights with a combination of creativity and analysis.
- Focus mindfully on the topic at hand.

4.2.3.3 ITIL continual improvement model

The ITIL continual improvement model provides organizations with a structured approach to implementing improvements.

Toyota Kata



Definition: Toyota Kata

A mental model and behaviour pattern for scientific thinking and routines for practice and coaching.

This four-step improvement approach is based on five questions:

- What are we trying to achieve?
- Where are we now?
- What obstacle is now in our way?
- What is our next step, and what do we expect?
- When can we see what we have learned from taking that step?

Application

When applying Toyota Kata, practitioners should take the following steps:

- **Understand the direction** Improvement should be aimed at specific goals, not just random.
- **Grasp the current condition** A direction is not useful unless we know where we are right now.
- **Establish the next target condition and identify obstacles** Describe both the outcome we desire next and the expected condition of the process to generate that outcome.
- **Experiment towards the next target condition** Come up with ideas to overcome an obstacle and run experiments with that idea. If possible, test only one hypothesis at a time.

OODA loop

The OODA loop is another improvement technique that can be used in the context of HVIT. OODA stands for 'observe, orient, decide, act'. The OODA loop originated in operations during military campaigns, and is now also applied in other types of organizations. The approach shows how agility can overcome raw power. It is especially applicable to cybersecurity, and is considered to make processes seem more immediately reactive than the more familiar PDCA (plan–do–check–act) approach. It is therefore popular in industries where unexpected challenges arise.

4.2.4 ITIL guiding principles

Some of the ITIL guiding principles manifest themselves more in an HVIT environment than others. This section outlines how each principle can be applied to an HVIT way of working.

4.2.4.1 Focus on value

The focus on value principle is reflected in two of the HVIT objectives: valuable investments and co-created value. This principle is particularly applicable to HVIT because the digital products and services that HVIT organizations provide will have an effect on their consumers. When working in an HVIT organization, it is important to keep in mind that value may need to be created for colleagues across the organization and for external stakeholders. Each stakeholder group might have a different expectation of value, and the organization needs to find ways to balance often conflicting needs. In an HVIT environment, stakeholders and their needs are likely to change often. As a result, the focus on value principle has a very strong connection with the principle of think and work holistically.

4.2.4.2 Start where you are

The start where you are principle is particularly applicable for an organization that is looking to undergo a digital transformation. The organization may already have many of the resources needed for high-velocity work, although it may need to modify them to meet new objectives and ways of working. There is a danger that, when pursuing an HVIT way of working, an organization may affect its existing relationships, wisdom, and intelligence. Although this may be necessary in some cases, organizations should be careful to review what they have before proceeding, as much of this may be kept for HVIT work. HVIT is about reducing the total time from idea to launch for the delivery of a

digital product. By considering options to re-use what is already in place rather than simply building from scratch, organizations can often deliver products to market faster.

4.2.4.3 Progress iteratively with feedback

Progressing iteratively with feedback is one of the core principles of the Toyota Kata way of improving, which offers guidance for step-by-step improvement based on data-driven, scientific thinking. An iterative approach is not limited to software development, and can be adopted by service management working in a high-velocity environment. Service delivery review cycles with key partners and vendors can be shortened from quarterly to monthly, or even weekly, cycles, creating a fast delivery and feedback loop to enable a high-velocity way of working.

4.2.4.4 Collaborate and promote visibility

This principle is particularly relevant to HVIT, as high-velocity work relies on quick access to different resources, including information, and a highly collaborative way of work. When working in an HVIT environment, it is important to:

- modulate the degree of independence, self-organization, collaboration, and visibility to minimize overhead work
- promote peer-to-peer visibility along a value stream, along with hierarchical visibility between senior management and leadership, and engineers and frontline staff.

4.2.4.5 Think and work holistically

Much thinking in HVIT rightly focuses on meeting the needs of consumers. However, there are many other stakeholders that can be positively, or negatively, impacted by high-velocity work. It is therefore important to think about the implications that decisions made by

high-velocity teams have for all stakeholders. For example, moving to cloud computing changes the balance of capital expenditure and revenue/operating expenditure, which in turn has an impact on financial governance and the results the organization reports to shareholders and tax authorities. Introducing new digital products and services can have knock-on effects on industries (famous examples include taxis, hospitality, and video rental), which in turn can create positive and negative outcomes for societies and local economies.

4.2.4.6 Keep it simple and practical

This principle applies to HVIT work and the systems surrounding and enabling it, and to the consumption of products and services. Keeping it simple does not mean keeping things simplistic, but refers to the philosophy of reducing overhead responsibilities, optimizing (or in some cases reducing) the degree of automation, and reducing friction between teams. An HVIT example is identifying the minimum information required to get work done, and not over-engineering monitoring and configuration management database (CMDB) tools to track everything that the organization can manage.

4.2.4.7 Optimize and automate

The optimize and automate principle manifests itself not only in the continual improvement of an organization's digital products and services, but also in the improvement of its processes. An HVIT example is documenting and streamlining the processes that are in place to test software before investing in CI/CD tools. It is a common mistake to optimize a single part of a system and to work for individual efficiency rather than overall effectiveness. This often creates queues that actually slow down the overall process, even when individual effectiveness is optimized.

5 High-velocity IT techniques

5.1 Techniques for valuable investments

See section 2.3.1 for a description of the valuable investment objective that the following techniques support.

5.1.1 Prioritization techniques

Queues occur wherever the demand for work exceeds the capacity to complete it within the expected timeframe. In an ideal situation, an organization would have no variation in demand and would have the appropriate quality and quantity of resources needed to satisfy it. However, organizations often need to contend with having a fixed capacity but a varying demand for services. This imbalance creates queues or backlogs in which work items need to be prioritized.

Prioritization is an activity commonly associated with support and software development work (for example, prioritizing incident investigation or a sprint backlog), but its use is universal.

5.1.1.1 Cost of delay

A useful technique for prioritization is estimating the cost of delay of a new or improved service offering. This refers to the financial and non-financial benefits that would be lost if a service activity or task were delayed. An understanding of the cost of delay gives practitioners the ability to prioritize work based on data on value, rather than their intuition. This applies to the initial prioritization and the continual evaluation and reprioritization of ongoing work within changing circumstances.

This technique is particularly useful in HVIT environments because investments are generally more significant and market conditions rapidly change, meaning it is important to continually assess options for alternative investments.

5.1.1.2 Buy/hold/sell

A portfolio of products (or other assets) can be managed using the buy/hold/sell technique. This involves assessing each product and deciding which of three investment strategies best applies to it:

- **Buy** Invest in improving or extending the product.
- **Hold** Spend as little as possible to maintain the product, as long as the costs are affordable.
- **Sell** Invest in retiring, reducing, or replacing the product.

This technique illuminates the difference between the costs and benefits of developing, maintaining, or retiring a product, along with the associated risks and trade-offs. It helps decision-makers to be explicit about their choices and to accept the consequences of their decisions.

5.1.1.3 Other techniques

There are many other product prioritization techniques that product/service owners can consider, including stacked ranking, Kano, net present value (NPV), return on investment (ROI), and fit/feasibility/attractiveness.

5.1.2 Minimum viable products and services

A minimum viable product or service is one that has just enough features to enable its early assessment and the collection of feedback for future development. The 'minimum viable' approach is an efficient way of developing products and services, particularly when the market is volatile and unpredictable. It is consistent with complexity thinking, which

recognizes that some things are unknowable and therefore that it is impossible to design a product or service with complete, predetermined requirements. When requirements are unknown, unarticulated, or ambiguous, experiments can establish what works and what doesn't. Therefore, a minimum viable approach enables valuable investments and contributes to fast development through an iterative working style.

A minimum viable product or service typically has three key characteristics:

- It has enough value that people are willing to use or buy it.
- It demonstrates the potential for enough benefit to retain those early adopters.
- It provides a feedback loop to guide future development.

5.1.2.1 Minimal marketable product

The next step after a minimum viable product is to develop something for a small group of initial users (innovators or early adopters). This is called a minimum marketable product.

5.1.3 Product or service ownership

Scrum suggests three roles: the product owner, the development team, and the Scrum master. The product owner is responsible for maximizing the value of the product that the development team produces. Product ownership entails establishing and prioritizing requirements, and communicating them to the development team. In the context of these software development teams, it is the product owner who liaises and negotiates with the various stakeholders, including consumers. HVIT environments are often product-oriented, so the concept of the product owner is highly relevant to HVIT. The concept of a product owner also applies to services and service owners.

Product or service ownership is a key part of all value chain activities, and contributes to valuable investments.

5.1.4 A/B testing

It can be difficult to predict whether a feature will be valuable to users. This problem could be solved by measuring user behaviour to gather solid data; however, when there are too many influencing factors, it is almost impossible to isolate the effect of the new feature. Therefore, a control group is required.

A/B testing is a time-limited experiment in which one group of users, the control group, is provided with an old version of a product or service. At the same time, another group of users, the treatment group, is provided with a new version of the product or service that includes the new feature. Assuming all other factors influencing both groups are equal, the measurements of the groups can be compared, thus gathering data for a value-based decision.

Because HVIT environments are often product-oriented and operate under time pressure in unpredictable markets, A/B testing is particularly relevant to HVIT.

5.2 ITIL practices contributing to valuable investments

Many ITIL 4 management practices contribute to HVIT and to achieving valuable investments. However, two key practices are outlined below. To read more about the practices, the practice guides are available on the AXELOS My ITIL platform (https://www.axelos.com/my-axelos/my-itil).

5.2.1 Portfolio management

5.2.1.1 Purpose

To ensure that the organization has the right mix of programmes, projects, products, and services to execute the organization's strategy within its funding and resource constraints.

5.2.1.2 Practice success factors

The portfolio management practice includes the following practice success factors:

- ensuring sound investment decisions for programmes, projects, products, and services within the organization's resource constraints
- ensuring the continual monitoring, review, and optimization of the organization's portfolios.

Table 5.1 outlines how some techniques have a high impact on this practice.

Table 5.1 Techniques that have high impact on portfolio management

Technique	Objective	Activities/resources
Prioritization techniques	Valuable investments	Continually prioritizing service offerings based on value, incorporating the cost of delay.
Minimum viable products and services		Using the concept of a minimal viable product as a dynamic decision-making tool to support good investments in the portfolio of features within a product or service.
Product or service ownership		Involvement of (software) product owners and product or service managers in evaluating and prioritizing the product or service investment proposals.
A/B testing		Deciding and prioritizing which services, products, and features to invest in using A/B testing data.
Continual business analysis	Fast development	Continually scanning customers, market conditions, and the broader ecosystem to understand their impact on the way the organization invests in various products and services.
		Monitoring ongoing portfolio investments to identify value leakage or to verify value co-creation.
Technical debt	Resilient operations	Deciding whether to invest resources to fix the technical debt present in live products and services, and understanding the impact on investment towards future products and services.
		Assessing technical debt so that new portfolio items can be introduced to the existing pool.

5.2.2 Relationship management

5.2.2.1 Purpose

To establish and nurture the links between the organization and its stakeholders at strategic and tactical levels. It includes the identification, analysis, monitoring, and continual improvement of relationships with and between stakeholders.

5.2.2.2 Practice success factors

The relationship management practice includes the following practice success factors:

- establishing and continually improving an effective approach to the relationship management practice across the organization
- ensuring effective and healthy relationships:
  - within the organization
  - between the organization and its external stakeholders.

Table 5.2 outlines how some techniques have a high impact on this practice.

Table 5.2 Techniques that have high impact on relationship management

Technique	Objective	Activities/resources
Product or service ownership	Valuable investments	Structuring interactions with stakeholders.
		Involvement in establishing customers' priorities for new or changed products and services.
		Coordinating customers' requirements and feedback.
		Involvement in addressing complaints and mediating conflicting requirements.
Continual business analysis	Fast development	Continually scanning customers, market conditions, and the broader ecosystem to understand their impacts on the organization's relationships with external stakeholders.

5.3 Techniques for fast development

See section 2.3.2 for a description of the fast development objective that the following techniques support.

5.3.1 Infrastructure as code

Infrastructure as code (IaC) enables faster provisioning of environments, contributing to faster development and more resilient operations. Virtualization and hypervisor technology (often provided via the cloud) allows infrastructure items to be created, modified, and removed remotely via programming interfaces. Today, it is common practice to build and configure servers using scripts and configuration files.

IaC is a way of managing and provisioning IT infrastructure and platforms by using machine-readable definition files rather than physically configuring hardware components. These files can then be

stored in a version control system (see version control in section 5.5.4). IaC is also more efficient than a traditional physical configuration because it has fewer repeatable manual operations, thereby contributing to valuable investments by reducing the costs of infrastructure.

The concept of idempotence is key to IaC. This means that a deployment command always configures the target environment into the specified state, regardless of what it was previously.

5.3.2 Loosely coupled information system architecture

Loosely coupled information system architecture is based on comparatively small, independent components. This architecture enables work to be done in small, independent product- or service-based teams and platform-based teams. By breaking a system down into parts that can be developed and managed separately, teams can focus on their own part and limit their interactions with other teams. The product- or service-based teams comprise developers and engineers, along with product/service owners, who represent the consumer perspective. This closer collaboration is beneficial for fast development, as well as co-creating value. It also contributes to valuable investments, fast development, and resilient operations (where IT operations are also represented in the team).

Techniques such as microservices and containerization are often associated with loosely coupled architecture.



Definitions

- **Microservices** A variation of the service-oriented architecture in which an application is designed and developed as a set of small, loosely coupled services, each running in its own process and using lightweight mechanisms to communicate.
- **Containerization** The technique of packaging software into standardized lightweight, stand-alone, executable units for development, shipment, and deployment.

The benefits of loosely coupled information system architecture include:

- It facilitates faster development and more frequent changes.
- It enables evolutionary architecture.
- It is more efficient, and leads to less work being needed on redesigning and redeveloping a product or service.
- It enables a switch from a team structure focused on components to a product- or feature-based team structure.

5.3.3 Reviews

Progressing iteratively with feedback implies having regular reviews of the achievements, to identify lessons to be learned and correct the course of action if necessary. However, these reviews should not slow down progress or introduce excessive controls.

5.3.3.1 Retrospectives

In Agile, a retrospective is a team meeting at the end of an iteration (or 'sprint') or project to discuss what went well, what could be improved,

and how to benefit from the findings in the future. This fast and frequent feedback contributes to fast development. Retrospectives can be applied to most scenarios, and in any area that needs continual improvement. To be effective, they should be scheduled regularly and a facilitator should smooth the contributions of participants. Actions that are input for continual improvement should be identified, and responsibilities delegated.

5.3.3.2 Blameless post-mortems

Digital technology has increasingly significant social and economic consequences, particularly in HVIT environments. It is therefore becoming more important to prevent incidents. However, complex systems are inherently hazardous: despite all efforts, they will fail. It is therefore also important to learn from failure.

A post-mortem is a formal record of an incident in terms of its impact, resolution/mitigation efforts, causes, and measures to prevent recurrence. The quality of a post-mortem is better when the participants feel able to share their knowledge and options without fear for their reputation and position. This is why it is crucial to focus on what, rather than who, caused an incident. This blameless culture originated in the healthcare and airline industries, where lives are at risk. Blameless post-mortems are strongly related to safety culture and psychological safety (see section 4.2.2.2).



Definition: Blameless post-mortem

A non-judgemental description and analysis of the circumstances and events that preceded an incident.

Post-mortems are not only useful for the people involved in the incident, but should be shared with and read by others who may benefit from the results. It is also important to share the findings with the broader community, including consumers, as transparency and vulnerability establish trust and help to co-create value.

Good behaviours relevant to post-mortems include:

- conducting post-mortems as part of 'business as usual', not as exceptions
- focusing on what caused an incident, not who
- sharing results broadly and transparently in order to learn and foster trust.

5.3.4 Continual business analysis

Rapidly changing, unpredictable HVIT environments require continual adjustments in response to fluctuating market demand. This has implications for business analysis.

When an investment decision has been made, it is crucial to verify any initial assumptions that were made about product or service specifications, characteristics, and features. Some providers ignore the need to interact with real users as soon as possible, and spend several months or even years on development before presenting their solution to customers and users. Often, when this approach is taken, some features of the product or service are unnecessary, some need significant readjustment, and others that would be highly valuable are missing. However, the organization's resources and time have already been spent, and marketing opportunities are shrinking.

A better way to develop is to implement feedback loops as early as possible. Proper feedback loops can provide valuable information about the direction in which the product or service should be developed. For a

feedback loop to be useful, it is important to develop iteratively. A minimum viable approach (see section 5.1.2) could be used for this purpose.

5.3.5 Continuous integration, continuous delivery, and continuous deployment

Continuous integration, continuous delivery, and continuous deployment (CI/CD) are descriptive terms for a collection of practices primarily associated with software engineering, which are central to the philosophy of Lean and Agile software development. The adoption of these practices has grown rapidly, and it is important to understand the defining characteristics of CI/CD and the wider context of evolving system development practices when implementing services that are underpinned by software development.



Definitions

- **Continuous integration** An approach to integrating, building, and testing code within the software development environment.
- **Continuous delivery** An approach to software development in which software can be released to production at any time. Frequent deployments are possible, but deployment decisions are taken case by case, usually because organizations prefer a slower rate of deployment.
- **Continuous deployment** An approach to software development in which changes go through the pipeline and are automatically put into the production environment, enabling multiple production deployments per day. Continuous deployment relies on continuous delivery.

CI/CD contributes to fast development and, because the deployments are more reliable, to resilient operations.

5.3.6 Continuous testing

Software testing is not just about testing software that is developed and operational. Testing is something that should be conducted throughout the entire software development lifecycle. Table 5.3 outlines the different types of testing.

Table 5.3 Types of software testing

Testing the ideas	All software originates with an idea, often an attempt to solve a problem. Testing the idea helps to determine its quality from a customer perspective (based on wants and needs) and a business perspective (based on metrics, including growth, revenue, conversion, and user base).
Testing the artefacts	Artefacts, such as epics, user stories, acceptance criteria, data flow diagrams, and process flow diagrams, should be tested. Inspecting the information within the artefacts can help to identify ambiguities, assumptions, and information relating to risks and variables. This information can be used as feedback to help review and improve the artefacts.
Testing the user experience and user interface designs	Artefacts are used for design activities, programming activities, and testing activities. The design activities produce more design artefacts than can be tested. Interfaces and experience are tested using relevant artefacts, and test results may affect other artefacts.
Testing the architecture and code designs	When designing software architecture and discussing how to build it and new features, exploratory testing can enhance the designs and ideas.

Testing the code	Code reviews are an important part of building a high-quality product. Anyone should be able to read the code and test it from different perspectives. Unit testing verifies that each unit of the software performs as it should. Unit tests are often automated. It is important to note that this is the first point within the software development lifecycle where there should be something tangible to measure against.
Testing the operational software	Testing the operational software is the most common activity that is associated with testing. Many Agile teams include 'testing' as a status within their workflow after development. However, testing should not be a status in a workflow; it should be a series of structured activities conducted throughout the software development lifecycle.
Testing in different environments	When it comes to test environments, a risk-based approach may be appropriate. There are many risks that can be tested for. Some of these cannot be tested for in a development environment. For others, a stringently integrated environment is needed for testing. Some risks can only be tested for in the real production environment.
Testing the release pipelines	Pipeline processes can be tested, and teams often do so implicitly in order to enhance their pipelines to be as efficient and fast as possible. This requires a good understanding of the structure behind pipeline testing.
Testing the system in production	Some risks must be tested for in the production environment, such as performance risks (particularly user load and user stress risks), user acceptance testing (where users use the system as they would the live software), and observability risks (to test the effectiveness of observability solutions and implementations).

Table continues

Table 5.3 *continued*

Testing the system in production *continued*	When testing in production, the aim should be to minimize the risk of testing affecting the customers. Strategies to achieve this include: ● **Canary releases** The new feature is initially released to a small, targeted group of users, gradually increasing to all users. ● **Feature toggles** Features can be hidden behind a feature flag, easily enabled or disabled by toggling the flag on or off. ● **Blue/green deployments** Two identical production environments run simultaneously (one 'blue' and one 'green'), with only one of the environments being live, serving all production traffic, whereas the other is used for deployment of the new versions. ● **Automated roll-back strategies** Automation tools can quickly revert the release to its previous version in the event of an incident.
Testing the services in production	There are many services, activities, and technologies that are implemented for released production software, which can all be tested. From a service delivery perspective, it is important to test processes. For example, when testing whether a user can access support, it is important to ask how easy it is for users to get support. What opportunities do users have within the software to do that?

To achieve continuous testing, a number of principles[5] should be in place, including:

- Tests should be written at the lowest level possible.
- Tests with the fewest external dependencies should be favoured.
- Write once, run anywhere, including in the production system.
- Products should be designed for testability.
- Test code is product code; only reliable tests survive.

- Testing infrastructure is a shared service.
- Test ownership follows product ownership.
- Fault injections and chaos engineering in production should be practised to test service resilience.
- Unreliable tests should be eliminated.

5.3.7 Kanban



Definition: Kanban

A Lean method based on a highly visualized pull-based workflow that manages and improves work across human systems by balancing demands with available capacity, and by improving the handling of system-level bottlenecks.

Kanban is a set of principles, practices, and regular activities aimed at developing and managing a predictable, rhythmic, and constant flow of work. When applied properly, it can significantly accelerate the development of high-quality products and services. The pull-based triggering mechanism allows customers to pull work through the value stream. Pull-based working also has the advantage that work is not pushed onto people, burdening them unnecessarily and unproductively. This is valuable in Lean teams, where overburden is a form of waste.

The main practices of Kanban are:

- visualizing work
- limiting work in progress
- managing flow

- making process policies explicit
- implementing feedback loops
- improving collaboration
- evolving experimentally.

Sometimes organizations only use Kanban boards to visualize work in progress. Although using a board is important, it is a limited application of Kanban. The power of Kanban relies upon holistic implementation and constant attention to the workflow.

Kanban suggests a system of regular meetings that ensure effective communication. This system is often referred to as 'Kanban cadences'. The frequency of cadences is specific to an organization and should be defined and adjusted case by case. The meetings are:

- strategy review
- operations review
- risk review
- service delivery review
- replenishment meeting
- delivery planning meeting.

5.4 ITIL practices contributing to fast development

Many ITIL 4 management practices contribute to HVIT and to achieving fast development. However, five key practices are outlined below. To read more about the practices, the practice guides are available on the AXELOS My ITIL platform (https://www.axelos.com/my-axelos/my-itil).

5.4.1 Architecture management

5.4.1.1 Purpose

To explain the different elements that form an organization. This practice explains how the elements interrelate to enable the organization to effectively achieve its current and future objectives. It provides the principles, standards, and tools that enable an organization to manage complex change in a structured and Agile way.

5.4.1.2 Practice success factors

The architecture management practice includes the following practice success factors:

- ensuring that the organization's strategy is supported with a target architecture
- ensuring that the organization's architecture is continually evolving to the target state.

Table 5.4 outlines how some techniques have a high impact on this practice.

Table 5.4 Techniques that have high impact on architecture management

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Using a minimum viable approach to describe the service, technical, information, or environmental architecture that will create the necessary constraints, boundaries, or enablers for other types of service or product work.
Infrastructure as code	Fast development	Verifying architectural decisions and comparing infrastructure solutions.
Loosely coupled information system architecture		Designing loosely coupled service, technical, and information architectures.
Continuous testing		Designing and improving service, technical, and information architecture to leverage CI/CD capabilities.

5.4.2 Business analysis

5.4.2.1 Purpose

To analyse a part or the entirety of a business, define its needs, and recommend solutions to address these needs and/or solve a business problem. The solutions must facilitate value creation for the stakeholders. Business analysis enables an organization to communicate its needs in a meaningful way and express the rationale for change. This practice enables an organization to design and describe solutions that enable value creation, in alignment with the organization's objectives.

5.4.2.2 Practice success factors

The business analysis practice includes the following practice success factors:

- establishing and continually improving an organization-wide approach to business analysis to ensure that it is conducted in a consistent and effective manner
- ensuring that current and future needs of the organization and its customers are understood, analysed, and supported with timely, efficient, and effective solution proposals.

Table 5.5 outlines how some techniques have a high impact on this practice.

Table 5.5 Techniques that have high impact on business analysis

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Using a minimum viable approach as a tool to extract core business value from a product or service.
Loosely coupled information system architecture	Fast development	Understanding consumer needs and translating them into detailed requirements for each component of a loosely coupled service architecture.
Continual business analysis		Continually scanning customers, market conditions, and the broader ecosystem to understand their impacts on the organization's products and services.
Service experience	Co-created value	Understanding user needs and translating them into customer experience or user experience requirements, in addition to traditional requirements regarding utility and warranty.

5.4.3 Deployment management

5.4.3.1 Purpose

To move new or changed hardware, software, documentation, processes, or any other component to live environments. It may also be involved in deploying components to other environments for testing or staging.

5.4.3.2 Continuous integration, continuous delivery, and continuous deployment (CI/CD)

The key concepts for deployment in Agile and DevOps are defined in section 5.3.5.

5.4.3.3 Practice success factors

The deployment management practice includes the following practice success factors:

- establishing and maintaining effective approaches to the deployment of services and service components across the organization
- ensuring the effective deployment of services and service components in the context of the organization's value streams.

Table 5.6 outlines how some techniques have a high impact on this practice.

Table 5.6 Techniques that have high impact on deployment management

Technique	Objective	Activities/resources
Infrastructure as code	Fast development	Automating the deployment of infrastructure, ensuring a faster and more repeatable and reliable deployment of both infrastructure and applications.
Loosely coupled information system architecture		The scope of deployments and deployment patterns is reduced with the decoupling of system architecture; this makes deployments easier to manage and replicate, and strong candidates for automation.
Blameless post-mortems		Investigating successful and failed deployments of service components to identify opportunities to improve the success of future deployments.
CI/CD		The systematic/automatic installation of specific versions or packages of software to a predetermined environment (integration, user acceptance testing, production).
Definition of done	Resilient operations	Deployment management activities can be structured around a definition of done; for example, to create a boundary with release management. When moving releases to live environments, teams should verify that deliverables for support are complete: all requirements, user stories, and tests should be accepted.
Version control		Using version-controlled repositories to deploy new or changed service components or return to a previous version.

5.4.4 Service validation and testing

5.4.4.1 Purpose

To ensure that new or changed products and services meet defined requirements. The definition of service value is based on input from customers, business objectives, and regulatory requirements and is documented as part of the design and transition value chain activity. These inputs are used to establish measurable quality and performance indicators that support the definition of assurance criteria and testing requirements.

5.4.4.2 Service validation

Service validation is performed in the earlier stages of the product and service lifecycle (ideation and design). It is focused on confirming that the proposed service design meets agreed service requirements and on establishing acceptance criteria for the next stages (development, deployment, and release). These criteria will then be verified by testing the product and service components, products, and services.

5.4.4.3 Testing

Based on the criteria identified through service validation, test strategies and test plans are developed and implemented. A test strategy defines an overall approach to testing. Test strategies can apply to environments, platforms, sets of services, and individual products or services. The product and service lifecycle stages that are covered by testing may differ between products and services developed within the organization and those obtained from a supplier.

5.4.4.4 Practice success factors

The service validation and testing practice includes the following practice success factors:

- defining and agreeing an approach to the validation and testing of the organization's products, services, and components in line with the organization's requirements for speed and quality of service changes
- ensuring that new and changed components, products, and services meet agreed criteria.

Table 5.7 outlines how some techniques have a high impact on this practice.

Table 5.7 Techniques that have high impact on service validation and testing

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Developing test cases to check that all service components support the minimum viable product or service.
Infrastructure as code	Fast development	Designing and maintaining test cases to ensure that virtual IaC meets organizational requirements and policies.
CI/CD		Developing automated test cases to support continual integration activities.
Continuous testing		Unit, integration, and regression testing is conducted on an ongoing basis throughout the development lifecycle. This includes application unit testing, infrastructure service testing, functional/non-functional testing, canary releases, blue/green testing, and infrastructure security testing.
Definition of done	Resilient operations	Testing activities can be structured around a definition of done to ensure that multiple types of tests are conducted.

5.4.5 Software development and management

5.4.5.1 Purpose

To ensure that applications meet internal and external stakeholder needs in terms of functionality, reliability, maintainability, compliance, and auditability.

5.4.5.2 Practice success factors

The software development and management practice includes the following practice success factors:

- agree and improve an organization's approach to development and management of software
- ensure that software continually meets an organization's requirements and quality criteria throughout its lifecycle.

Table 5.8 outlines how some techniques have a high impact on this practice.



Definitions (abbreviated)

Definition of done A checklist of the agreed criteria for a proposed product or service.

Maintenance The modification of the application as part of development, for both correction and enhancement purposes: corrective, preventive, adaptive, and perfective.

Scrum® An iterative, timeboxed approach to product delivery that is described as 'a framework within which people can address complex adaptive problems, while productively and creatively delivering products of the highest possible value'.

Software A set of instructions that tell the physical components (hardware) of a computer how to work.

Software development The design and construction of applications according to functional and non-functional requirements, and correction and enhancement of operational application according to changing functional and non-functional requirements.

Software development lifecycle (SDLC) model The sequence in which the stages of the software development lifecycle are executed. The major stages are: establish requirements; design; code; test; run/use the application.

Software quality A qualification of the value of software as a product and in its use, such as the one categorized by ISO/IEC 25010:2011.

Technical debt The total rework backlog accumulated by choosing workarounds instead of system solutions that would take longer.

Table 5.8 Techniques that have high impact on software development and management

Technique	Objective	Activities/resources
Prioritization techniques	Valuable investments	Calculating the financial impact of delaying work on new software features or larger software-based service components.
		Deciding whether to obtain or build software components.
Minimum viable products and services		Using a minimum viable approach as a decision-making tool to prioritize work on software features.
Product or service ownership		Involvement of product and service owners in articulating, refining, and prioritizing the software development backlog, and in deciding whether to acquire or upgrade commercially available software.
Infrastructure as code	Fast development	Developing software architecture and code to leverage fast delivery/provisioning of virtual hardware infrastructure.
Loosely coupled information system architecture		The detailed design, building, running, and management of loosely coupled software components.
Retrospectives		Activities and techniques to review work periodically to understand how to improve.
Continual business analysis		Continually scanning customers, market conditions, and the broader ecosystem to understand their impacts on the organization's software products and the prioritization of ongoing software development work.
CI/CD		Use of CI/CD techniques that are used for application software.
Kanban		Visualizing and improving the flow of new or changed software components into live environments.

Technique	Objective	Activities/resources
Technical debt	Resilient operations	Identifying and reducing technical debt by creating or modifying infrastructure and platform service components.
Chaos engineering		Chaos engineering tools are themselves software applications that need to be developed (or configured) and managed. Software should be designed and architected with sufficient resilience and redundancy.
Definition of done		Software can be developed (or configured) to meet a definition of done before it is deployed into live environments, ensuring that code is understandable, maintainable, and ready to support future changes.
Version control		Code, and even configuration settings for other software components, can be formally managed using a version-controlled repository to house the outputs of software development and management work.
Site reliability engineering		Providing requirements to SRE teams and acting on feedback.
Service experience	Co-created value	The desired service experience informs the design of the user interface.
Peer review	Assured conformance	Inspecting development work between peers to increase the quality of code to ensure that it effectively satisfies demand and performance expectations.

5.5 Techniques for resilient operations

See section 2.3.3 for a description of the resilient operations objective that the following techniques support.

5.5.1 Technical debt

In software development and management, technical debt is the total backlog of rework needed to repair substandard (changes to) software. Generally, when software is used, enhancements and repairs will follow. When these changes are applied, the quality of the software will decline unless effort is made to limit the damage. This is known as 'software entropy', which is particularly relevant to HVIT because of the large investments in software, the frequency of changes, and the need to change quickly to keep up with market demand.

Technical debt is also defined as the backlog of work that is needed to fix damage that is incurred while enhancing or repairing the software. Such damage is usually caused by a combination of limited time, money, knowledge, and skills.

Reducing technical debt results in fewer incidents, and therefore contributes to resilient operations. It also contributes to valuable investments and fast development. A balance between these benefits should be achieved and maintained. It is therefore prudent to recognize the existence of technical debt, qualify and quantify it as best possible, assess the associated short-term and long-term risks, and make appropriate decisions as part of (software) product management, including allocating a budget to 'repay' the technical debt.

5.5.2 Chaos engineering



Definition: Chaos engineering

The discipline of experimenting on a system in order to build confidence in the system's capability to withstand turbulent conditions in production.

The principles for an ideal application of chaos engineering are:

- **Build and hypothesize around steady state behaviour** Focus on measurable system outputs, rather than attributes of the system.
- **Vary real-world events** Chaos variables reflect real-world events. Prioritization must be defined by considering the impact or estimated frequency of the events.
- **Run experiments in production** Chaos prefers to experiment directly on production traffic.
- **Automate experiments to run continuously** Build automation into the systems to drive orchestration and analysis.
- **Minimize the blast radius** The fallout from experiments should be minimized and contained.

Chaos monkey is one of the tools in the so-called Simian army,[6] the best-known toolset for chaos engineering. The Simian army is a collection of open-source cloud testing tools created by Netflix®. Chaos monkey tests responses to system failures by disabling certain components to see how the remaining systems will respond. Although chaos monkey may disrupt operations, it also contributes to their longer-term resilience.



Definition: Chaos monkey

A tool that tests the resilience of IT systems by intentionally disabling components in production to test how remaining systems respond to the outage.

Deployment of the chaos monkey results in the termination of a randomly selected system within an identified group of systems. This creates a situation close to a natural incident and tests the ability of the systems to sustain failures.

5.5.3 Definition of done

The 'definition of done' (see section 5.4.5.1) is commonly used by application developers employing the Agile Scrum methodology. The Scrum community defines 'done' as 'having produced potentially releasable software increments'. In DevOps circles, this has been extended to include 'released to production'. This much-needed extension is still inadequate when seen from an end-to-end perspective. Releasing the required functionality is not useful if users use it poorly, interpret data incorrectly, make sub-optimal decisions, or fail to act on good decisions. A narrowly focused IT function may not see preventing this as its responsibility, but as the organization's responsibility. Therefore, a more complete definition of done includes criteria for operations and use. The technique is broadly applicable and is often encountered in HVIT environments.

A definition of done describes the functional criteria that contribute to a product's or service's utility, and the non-functional criteria that contribute to its warranty. These non-functional criteria should be defined and agreed with whoever is responsible for operations. A definition of done that includes non-functional criteria therefore contributes to resilient operations and to value co-created through improved usability, and also to faster development because less rework is needed.

5.5.4 Version control



Definition: Version control

The administrative management of sources and artefacts of information systems, products, and services.

Version control is of particular relevance to HVIT, because a strong correlation has been established between good version control and high IT performance: it provides a quicker lead time for change, more frequent deployment, and a quicker mean time to restore service.

Version control tracks which versions of sources and artefacts exist in which environments. The common practice of storing software source code in a version control system can be expanded to include almost every other significant component of an IT system, product, or service. Version control therefore applies not only to service components (developed/engineered and managed in the obtain/build value chain activity), but also at the level of services.

Version control contributes to resilient operations and fast development. It can also be considered as an enabler for evolutionary architecture: a practice in which architectural decisions are not restrictive, but rather enable the continual evolution and improvement of products and services without redesigning and redeveloping previous solutions.

5.5.5 AIOps



Definition: AIOps

The application of machine learning and big data to IT operations to receive continuous insights which provide continuous fixes and improvements via automation. Also referred to as 'artificial intelligence for IT operations' or 'algorithmic IT operations'.

AIOps aims to bring artificial intelligence (AI) to IT operations, addressing the challenges posed by modern trends in the ongoing evolution of infrastructure, such as the growth of software-defined systems. The implications of these new technologies, such as the increase in the rate at which infrastructure is reconfigured and reshaped, necessitates more automated and dynamic management technologies, which may have a significant impact on an organization's digital services.

AIOps platforms are used to enhance and partially replace many primary IT operations functions, such as availability and performance monitoring, failure recognition, predictive analysis, and event correlation and analysis.

AIOps harnesses data platforms and machine learning, collecting observational data (for example, events, log files, operating metrics) and engagement data (for example, customer request and service desk tickets), and drawing insights by applying cognitive or algorithmic processing to this data. The application of AIOps depends heavily on the availability of data to be analysed and whether the data lends itself for analysis, which it may not if it has an extremely complex nature and a weak correlation between symptoms, causes, and effects.

5.5.6 ChatOps

ChatOps is a model in which people, tools, process, and automation are connected in a transparent flow. This model helps to control pipelines and collaboration. It is a close-fitting integration of instant communications with operational execution: an emerging movement that promotes the integration of several teams, tools, and DevOps platforms. Development is driven by bringing tools and platforms into conversations. When bots are team members, it is possible to send requests to them and get instant responses.

ChatOps enables collaborative communication among humans and tools, reducing incident response time by eliminating repetitive requests for information and automating some regular IT operations actions. The elements of ChatOps include:

- **Chat platform** The service that connects stakeholders, teams, and the systems they work on.
- **Bots** The core of the ChatOps model. Bots exist and work between the collaboration tool and DevOps tools. They receive requests from team members, then retrieve information from the integrated systems by executing scripts.
- **Integration and automation services** The third-party elements in ChatOps; for example, for issue tracking: version control systems, IaC, continuous integration servers, or monitoring tools.

ChatOps is the evolution of traditional chats, as it enables systems to join the conversation. For example, DevOps or IT and service management tools can notify support groups of an incident or event.[7]

5.5.7 Site reliability engineering



Definition: Site reliability engineering

A discipline that incorporates aspects of software engineering and applies them to infrastructure and operations problems with the goal of creating ultra-scalable and highly reliable software systems.

Because highly digitally enabled organizations demand highly resilient operations, site reliability engineering (SRE) is particularly relevant to HVIT.

SRE applies a software development mindset to IT operations, and helps to align development and operations. SRE teams split their time between executing IT operations, coaching IT operations teams, and developing software that increases the resilience and performance of IT systems. They tend to spend less than half of their time on toil (otherwise, this indicates a problematic system). The more work is manual, repetitive, automatable, tactical, devoid of enduring value, and scales linearly as a service grows, the more likely it is to be toil.

SRE is based on experience that suggests that systems are complex and will therefore fail. Effort is therefore balanced between preventing failures and reducing the impacts of unpreventable failures, such as by designing systems to degrade gradually rather than fail abruptly. Failure is regarded as an opportunity to learn, recognizing that people are the adaptable element of complex systems, and that their expertise is changing just as much as the other parts of the system.

The improvements that SRE brings to availability, latency, and performance all contribute to resilient operations.

5.6 ITIL practices contributing to resilient operations

Many ITIL 4 management practices contribute to HVIT and to achieving resilient operations. However, six key practices are outlined below. To read more about the practices, the practice guides are available on the AXELOS My ITIL platform (https://www.axelos.com/my-axelos/my-itil).

5.6.1 Availability management

5.6.1.1 Purpose

To ensure that services deliver the agreed levels of availability to meet the needs of customers and users.

5.6.1.2 Practice success factors

The availability management practice includes the following practice success factors:

- identifying service availability requirements
- measuring, assessing, and reporting service availability
- treating service availability risks.

Table 5.9 outlines how some techniques have a high impact on this practice.

Table 5.9 Techniques that have high impact on availability management

Technique	Objective	Activities/resources
Definition of done	Resilient operations	Detailed warranty requirements for the new or changed service should be negotiated and agreed with stakeholders.
Site reliability engineering		Using SRE techniques and tooling to improve visibility of a system in order to judge the service health and diagnose issues.
		Tracking the 'technical' mean time between failures (MTBF) and (more crucially) mean time to restore service (MTRS) metrics, such as user outage minutes, number of lost transactions, lost business value, and user satisfaction.
		Using error budgets to balance service reliability and innovation.

5.6.2 Capacity and performance management

5.6.2.1 Purpose

To ensure that services achieve the agreed and expected levels of performance and satisfy current and future demand in a cost-effective way.

5.6.2.2 Practice success factors

The capacity and performance management practice includes the following practice success factors:

- identifying service capacity and performance requirements
- measuring, assessing, and reporting service performance and capacity
- treating service performance and capacity risks.

Table 5.10 outlines how some techniques have a high impact on this practice.

Table 5.10 Techniques that have high impact on capacity and performance management

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Using capacity and performance management as a basis for calculating the minimum resources (number of servers, number of service desk agents, etc.) required for a minimum viable product or service.
Definition of done	Resilient operations	A definition of done checklist must consider capacity requirements, demand forecasting, and performance for managing business and customer expectations.
AIOps		AIOps provides capabilities for identifying patterns and anomalies, determining the capacity and utilization of assets, and planning the capacity of future products or services.
Site reliability engineering		Using SRE techniques and tooling to improve visibility of a system in order to judge the service health and diagnose issues.
		Monitoring systems and defined service-level objectives (SLOs) must be accounted for and measured.
		Improving monitoring in order to better understand the system when things go wrong.

5.6.3 Infrastructure and platform management

5.6.3.1 Purpose

To oversee the infrastructure and platforms used by an organization. When carried out properly, this practice enables the monitoring of technology solutions available to the organization, including the technology and external service providers.

5.6.3.2 Practice success factors

The infrastructure and platform management practice includes the following practice success factors:

- establishing an infrastructure and platform management approach to meet evolving organizational needs
- ensuring that the infrastructure and platform solutions meet the organization's current and anticipated needs.

Table 5.11 outlines how some techniques have a high impact on this practice.

Table 5.11 Techniques that have high impact on infrastructure and platform management

Technique	Objective	Activities/resources
Product or service ownership	Valuable investments	Involvement of product and service owners in articulating, refining, and prioritizing the infrastructure and platform development backlog(s), and in deciding whether to acquire commercially available infrastructure components and services.

Technique	Objective	Activities/resources
Loosely coupled information system architecture	Fast development	The detailed design, building, running, and management of loosely coupled infrastructure and platform components.
Continual business analysis		Continually scanning customers, market conditions, and the broader ecosystem to understand their impacts on the organization's infrastructure and platform products.
CI/CD		Use of CI/CD techniques for digital infrastructure.
Technical debt	Resilient operations	Identifying and reducing technical debt by creating or modifying infrastructure and platform service components.
Chaos engineering		Designing infrastructure and platforms for sufficient resilience and redundancy to deal with the unexpected outages caused by chaos engineering tools.
		Providing information for chaos engineering regarding service components and backup activities.
Version control		Infrastructure components, configuration settings, and virtual and physical infrastructure components can be formally stored and managed using a version-controlled repository.
AIOps		AIOps tools can automate much of the day-to-day management of infrastructure and platform resources.
Site reliability engineering		Using SRE techniques and tooling to help architect and design infrastructure and platform capabilities to meet the organization's needs.

5.6.4 Monitoring and event management

5.6.4.1 Purpose

To systematically observe services and service components, and record and report selected changes of state identified as events. This practice identifies and prioritizes infrastructure, services, business processes, and information security events, and establishes the appropriate response to those events, including responding to conditions that could lead to potential faults or incidents.

5.6.4.2 Practice success factors

The monitoring and event management practice includes the following practice success factors:

- establishing and maintaining approaches/models that describe the various types of events and monitoring capabilities needed to detect them
- ensuring that timely, relevant, and sufficient monitoring data is available to relevant stakeholders
- ensuring that events are detected, interpreted, and if needed acted upon as quickly as possible.

Table 5.12 outlines how some techniques have a high impact on this practice.

Table 5.12 Techniques that have high impact on monitoring and event management

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Using monitoring and event management as the basis for designing and configuring tools for monitoring and telemetry, which are used to operate and learn from the minimum viable product or service.
AIOps	Resilient operations	AIOps tools can help to correlate vast data sets from across multiple monitoring tools. They create a better understanding of the IT environment.
		AIOps enables value co-creation through an integrated set of business and operational metrics, thereby reducing the frequency of operational events or incidents because they are predicted and prevented.
		AIOps helps to optimize IT and reduce IT costs by replacing silo-focused IT monitoring tools, and by monitoring the health and performance of applications of all tiers in value streams.
Site reliability engineering		Using SRE techniques and tooling to improve the visibility of a system in order to judge the service health and diagnose issues.
DevOps Audit Defense Toolkit	Assured conformance	Operational data warehouses consolidating performance and event data provide a rich repository of information to audit implementation and performance of controls.
DevSecOps		Configuring monitoring tools to continually scan for threats and vulnerabilities so that they can be escalated to the appropriate teams.

5.6.5 Problem management

5.6.5.1 Purpose

To reduce the likelihood and impact of incidents by identifying actual and potential causes of incidents, and managing workarounds and known errors.

5.6.5.2 Practice success factors

The problem management practice includes the following practice success factors:

- identifying and understanding the problems and their impact on services
- optimizing problem resolution and mitigation.

Table 5.13 outlines how some techniques have a high impact on this practice.

Table 5.13 Techniques that have high impact on problem management

Technique	Objective	Activities/resources
Prioritization techniques	Valuable investments	Calculating the financial cost of open problems and errors in order to prioritize and direct problem management efforts.
		Comparing the costs of workarounds with those of longer-term solutions.

Technique	Objective	Activities/resources
Loosely coupled information system architecture	Fast development	Investigating problems and designing error controls that span multiple loosely coupled systems.
Blameless post-mortems		Adjusting the leadership approach to examine the system rather than the people.
		Blameless post-mortems help organizations to obtain more information about the circumstances related to incidents. This provides better information for problem identification and investigation.
Technical debt	Resilient operations	Applying problem control and error control methods to manage technical debt.
Site reliability engineering		Data from SRE tools can help to identify problems, ensuring workarounds are applied quickly through the use of automation.
		Automating IT processes improves resilience and reduces toil.
		Use of post-mortems.

5.6.6 Service continuity management

5.6.6.1 Purpose

To ensure that the availability and performance of a service are maintained at sufficient levels in case of a disaster. The practice provides a framework for building organizational resilience with the capability of producing an effective response that safeguards the

interests of key stakeholders and the organization's reputation, brand, and value-creating activities.

5.6.6.2 Practice success factors

The service continuity management practice includes the following practice success factors:

- developing and managing service continuity plans
- mitigating service continuity risks
- ensuring awareness and readiness.

Table 5.14 outlines how some techniques have a high impact on this practice.

Table 5.14 Techniques that have high impact on service continuity management

Technique	Objective	Activities/resources
Chaos engineering	Resilient operations	Designing service continuity measures with sufficient resilience and redundancy to cope with the unexpected outages caused by chaos engineering tools.
		Continually monitoring continuity plans, measures, and mechanisms for resilience.

5.7 Techniques for co-created value

See section 2.3.4 for a description of the co-created value objective that the following technique supports.

5.7.1 Service experience

'Service experience' refers to the fact that service consumers value a service that is based on a combination of the 'technical' output of the service and how it is perceived from a human perspective. This means that service providers should be increasingly aware of consumer requirements and the resources that they have at their disposal to co-create value. Services are not passively received: value co-creation requires effort from the consumer. The service provider and consumer have to dynamically respond to each other's behaviour and accommodate exceptions as much as possible.

When, in some digitally enabled organizations, business and IT converge into a single organizational entity, there are no longer business and IT entities that need to be aligned. There is, therefore, no longer a business–IT relationship to manage. 'Business people' and 'IT people' report to the same management, have the same targets, and are often physically co-located.

Data analytics and machine learning can contribute significantly to relationship management. Information security is also important, as are ethics. Customer experience management and customer journeys are other topics that deserve consideration.

For further details on service relationships and service experience, see *ITIL®4: Drive Stakeholder Value*.

5.8 ITIL practices contributing to co-created value

Many ITIL 4 management practices contribute to HVIT and to achieving co-created value. However, three key practices are outlined below. To read more about the practices, the practice guides are available on the AXELOS My ITIL platform (https://www.axelos.com/my-axelos/my-itil).

5.8.1 Relationship management

For purpose, practice success factors, and techniques that have a high impact, see section 5.2.2.

5.8.2 Service design

5.8.2.1 Purpose

To design products and services that are fit for purpose and use, and that can be delivered by the organization and its ecosystem. This includes planning and organizing people, partners and suppliers, information, communication, technology, and practices for new or changed products and services, and the interaction between the organization and its customers.

5.8.2.2 Practice success factors

The service design practice includes the following practice success factors:

- establishing and maintaining an effective organization-wide approach to service design
- ensuring that services are fit for purpose and fit for use throughout their lifecycle.

Table 5.15 outlines how some techniques have a high impact on this practice.

Table 5.15 Techniques that have high impact on service design

Technique	Objective	Activities/resources
Minimum viable products and services	Valuable investments	Using a minimum viable approach to design the necessary customer experience and user experience elements of a product or service.
A/B testing		Using A/B testing methods to determine the effectiveness of customer experience and user experience prototypes before making further investments and design decisions.
Continual business analysis	Fast development	Continually scanning customers, market conditions, and the broader ecosystem to understand their impacts on the customers' and users' experiences of the organization's products and services.
Site reliability engineering	Resilient operations	SRE collaboration during the design phase can prevent a variety of problems or incidents from occurring later in production. Although design decisions can be reversed or rectified later in the development lifecycle, such changes come at a high cost in terms of effort and complexity.
Service experience	Co-created value	Articulating customer experience and user experience needs beyond a basic experience.

5.8.3 Service desk

5.8.3.1 Purpose

To capture demand for incident resolution and service requests. It should also be the entry point and single point of contact for the service provider for all users.

5.8.3.2 Practice success factors

The service desk practice includes the following practice success factors:

- enabling and continually improving effective, efficient, and convenient communications between the service provider and its users
- enabling the effective integration of user communications into value streams.

Table 5.16 outlines how some techniques have a high impact on this practice.

Table 5.16 Techniques that have high impact on service desk

Technique	Objective	Activities/resources
ChatOps	Resilient operations	Communicating and coordinating with users to better manage incidents and requests.
Service experience	Co-created value	Being empathetic and having the emotional intelligence to understand users' experiential needs.

Technique	Objective	Activities/resources
Service experience *continued*	Co-created value *continued*	Giving users a choice of communication channels.
		Service experience requires technology and information enablers, such as self-service tools, online portals, mobile applications, call centre tools, and chat.
		Using user satisfaction as a key performance indicator (KPI).
		Assessing user experience, while choosing a tool for two-way communication with users.
		Gathering service experience data (rough estimates of users happy/not happy with the service).

5.9 Techniques for assured conformance

See section 2.3.5 for a description of the assured conformance objective that the following techniques support.

5.9.1 DevOps Audit Defense Toolkit

The DevOps Audit Defense Toolkit[8] is guidance that addresses the tension between the IT function and auditors that is caused by new, more fluid patterns of work found in the DevOps community. It helps to demonstrate to auditors that the IT function understands the business risks and is properly mitigating them. By suggesting techniques to mitigate risk and to create a common perspective and shared understanding between the IT function and auditors, the toolkit contributes to assured conformance. By reducing unnecessary bureaucracy, it also contributes to fast development.

The DevOps Audit Defense Toolkit is relevant to HVIT because some of HVIT's principles and techniques appear to contradict conventional compliance requirements. Usually, however, this is a case of finding other ways of achieving the desired results. Internal regulations are derived from external requirements, and alternative internal regulations can often be found. It is essential, however, to involve auditors in this process.

5.9.2 DevSecOps

Most organizations have a dedicated information security team, who perform risk assessments and define policies, procedures, and controls. In high-velocity environments, information security is integrated as much as possible into the daily work of development and operations, shifting the reliance on control of process towards verifying preconditions, such as employees' expertise and integrity. The security officer's role shifts from 'policing' to enabling others to take necessary measures.

'DevSecOps' refers to this integration of security-related activities into the daily work of application development and IT operations. Security is built into the entire DevOps process and across the four pillars of culture, automation, metrics, and sharing (CAMS, or CALMS with the addition of Lean).



Definition: Integration of duties

Having a task that is prone to fraud or error performed by one person because other controls have been applied. This serves as an alternative to separation (or segregation) of duties.

Traditionally, duties are separated in order to reduce the risk of fraud and error; for example, the risk that untested and unauthorized code will be deployed into production. This can, however, lead to delay, and frustration at the perceived bureaucracy. The separation of duties is not an objective in itself: it is a method of reaching an objective. Other means are available to achieve the same objective, so duties can be integrated while maintaining the same level of assurance.

5.9.3 Peer review



Definition: Peer review

A judgement on a piece of scientific or other professional work by others working in the same area. When applied in software development, a work product is examined by its developer and one or more colleagues in order to evaluate its technical content and quality. This contributes to assured conformance.

Based on the demonstrated value of peer reviews in the engineering industry, numerous industry experts have identified it as a very desirable development practice. Experience has shown that problems (defects) are eliminated earlier if a development process incorporates peer reviews; these reviews are as effective as, or even more effective than, testing.[9]

Peer review provides a disciplined engineering practice for detecting and correcting defects in design artefacts. It has also been found to be one of the most effective ways of promoting the quality and productivity of design processes in software engineering and other engineering disciplines, including electrical, civil, mechanical, and fire protection engineering.

Data that is collected during the peer review process is used to correct defects, and to evaluate and improve the development process itself.

5.10 ITIL practices contributing to assured conformance

Many ITIL 4 management practices contribute to HVIT and to achieving assured conformance. However, two key practices are outlined below. To read more about the practices, the practice guides are available on the AXELOS My ITIL platform (https://www.axelos.com/my-axelos/my-itil).

5.10.1 Information security management

5.10.1.1 Purpose

To protect the information needed by the organization to conduct its business. This includes understanding and managing risks to the confidentiality, integrity, and availability of information, as well as other aspects of information security such as authentication and non-repudiation.

5.10.1.2 Practice success factors

The information security management practice includes the following practice success factors:

- developing and managing information security policies and plans
- mitigating information security risks
- exercising and testing information security management plans
- embedding information security into all aspects of the service value system.

Table 5.17 outlines how some techniques have a high impact on this practice.

Table 5.17 Techniques that have high impact on information security management

Technique	Objective	Activities/resources
Infrastructure as code	Fast development	Designing and enforcing security policies in virtual infrastructure components.
Loosely coupled information system architecture		Designing and managing information security policies for loosely coupled services and service components.
Definition of done	Resilient operations	Security tests such as vulnerability, penetration, or policy compliance should be considered in the definition of done for resilient products and services.
Version control		Addressing or closing information security risks by flagging vulnerable versions of service components.
DevOps Audit Defense Toolkit	Assured conformance	Designing and implementing controls in the product lifecycle to provide extensive traceability and joint accountability.
DevSecOps		Designing and implementing controls in a development lifecycle to provide extensive traceability and joint accountability.
		Integrating information security duties into the daily work of practitioners.

5.10.2 Risk management

5.10.2.1 Purpose

To ensure that the organization understands and effectively handles risks. Managing risk is essential to ensuring the ongoing sustainability of an organization and co-creating value for its customers. Risk

management is an integral part of all organizational activities and therefore central to the organization's service value system (SVS).

5.10.2.2 Practice success factors

The risk management practice includes the following practice success factors:

- establishing governance of risk management
- nurturing a risk management culture and identifying risks
- analysing and evaluating risks
- treating, monitoring, and reviewing risks.

Table 5.18 outlines how some techniques have a high impact on this practice.

Table 5.18 Techniques that have high impact on risk management

Technique	Objective	Activities/resources
A/B testing	Valuable investments	Using A/B testing methods to determine the effectiveness of risk mitigation options before making further investments.
Continual business analysis	Fast development	Continually scanning internal and external corporate risks to assess and understand their impact on the organization's products and services, and to design appropriate mitigations and countermeasures.
Peer review	Assured conformance	Reducing the risk of an unauthorized change being developed and released into production.
		Cross-checking between identification and assessment of risks.

6 Taking the HVIT examination

6.1 Purpose of the ITIL 4 HVIT qualification

The purpose of the ITIL 4 HVIT qualification is to provide the candidate with an understanding of the ways in which digital organizations and digital operating models function in high-velocity environments, focusing on the rapid delivery of products and services to obtain maximum value. The qualification will provide the candidate with an understanding of working practices such as Agile and Lean, and technical practices and technologies such as the cloud, automation, and automatic testing.

The ITIL 4 HVIT qualification is one of the prerequisites for the designation of ITIL 4 Managing Professional which assesses the candidate's practical and technical knowledge about how to run successful, modern IT-enabled services, teams, and workflows.

6.2 Examination overview

- **Material allowed** No additional materials are allowed in the examination. This is a 'closed book' examination.
- **Examination duration** The examination is scheduled for 90 minutes. Candidates taking the examination in a language that is not their native tongue or working language may be awarded 25% extra time, i.e. 113 minutes in total. The training or examination provider should be made aware of this when the course is booked.
- **Prerequisites** The candidate must have passed the ITIL 4 Foundation examination. In addition, the candidate must have attended an accredited training course for this module (the recommended duration for this training is 18 hours, including the examination).

- **Scoring** Number of questions: 40, each worth 1 mark (40 marks in total). There is no negative marking.
- **Provisional pass mark** 70% or higher – a score of 28 marks or above.
- **Level of thinking** The examination conforms to a learning objective tool called Bloom's Taxonomy. The term 'Bloom's levels' indicates the type of thinking needed to answer the question. The examination is constructed at Bloom's Level 2 (13 questions) and Bloom's Level 3 (27 questions). Weightings by learning outcome are as shown in Table 6.1.

Table 6.1 Examination weighting by learning outcome

Learning outcome	Number of questions	Approx. weighting (%)
Understand concepts regarding the high-velocity nature of the digital enterprise, including the demand it places on IT	7	17.5
Understand the digital product lifecycle in terms of the ITIL operating model	3	7.5
Understand the importance of the ITIL guiding principles and other fundamental concepts for delivering high-velocity IT	6	15
Know how to contribute to achieving value with digital products	24	60

6.3 Question type examples

There are three main question types: standard, negative, and list.

6.3.1 Standard

Which is a source of best practice?

(a) P

(b) Q

(c) R

(d) S

Study can be aided by trying to recall what the wrong answers are referring to.

6.3.2 Negative

Which is NOT a defined area of value?

(a) P

(b) Q

(c) R

(d) S

'Negative' questions are a type of 'standard' question in which the stem is negatively worded. These are traditionally the most difficult to answer, because, under the pressure of an examination, it is easy for the candidate to convince themselves that all options are true.

6.3.3 List

Which statement about service asset and configuration management is CORRECT?

1 It does P

2 It does Q

3 It does R

4 It does S

(a) 1 and 2

(b) 2 and 3

(c) 3 and 4

(d) 1 and 4

For a 'list' question, there is a list of four statements, and candidates have to select two correct ones. List-style questions are never negative.* Candidates should approach these questions by working out which statements they are certain are correct or incorrect. Knowing just two of the four will always lead to the correct answer.

6.4 Examination modalities

The examination can be taken as a paper-based exercise or in an online digital format. More information on examination regulations and technical requirements is available from the organizations providing training or examinations.

6.4.1 Sample papers

Two official sample examination papers will be provided as part of an accredited course.

These papers are written by the same examining teams as the live papers, and follow the same algorithm regarding the numbers and styles of questions. They also contain examiners' rationales which justify the correct answers.

* Negative questions are only used as an exception, where part of the learning outcome is to know that something is not done or should not occur.

Some training organizations will provide extra questions written by their own people. Care should be taken with these questions, as they are not always written to the same style and quality guidelines as those in the official sample and live papers. Nevertheless, they can prove helpful to aid a candidate's learning.

6.4.2 Tips for taking the examination

- **Read the question and highlight key words** It is very easy to miss a word in a question. For example, subconsciously removing 'NOT' from a question changes the focus completely. The human brain has a prediction mechanism that allows us to anticipate words without having to read them. Be careful to read the question fully and highlight key words to avoid mis-reading.
- **Be careful with time management, but don't stress about it** Forty questions in 90 minutes is 2 minutes and 15 seconds per question. Based on extensive experience, it is very unusual for time to be an issue. This is the case even for candidates whose native language is not English (who are given extra time). It is one less thing to worry about.
- **Remember 'What would ITIL recommend?'** The recommended duration for this training is 18 hours, including the examination. When answering questions, a candidate's first thought will be from short-term memory and faithful to what ITIL would say. Care should be taken not to rationalize questions against a candidate's experience of their own organization. The candidate should bear in mind that their organization will have adapted ITIL heavily, and its ways of working may not match exactly what the ITIL 4 Managing Professional Specialist titles say!

7 The ITIL 4 certification scheme

There are four levels within the ITIL 4 certification scheme, as shown in Figure 7.1.

7.1 ITIL Foundation

The ITIL 4 Foundation certification is designed as an introduction to ITIL 4 and enables candidates to look at IT service management through an end-to-end operating model for the creation, delivery, and continual improvement of IT-enabled products and services.

The target audience consists of:

- those who require a basic understanding of the ITIL framework
- those who want to understand how ITIL can be used to enhance IT service management

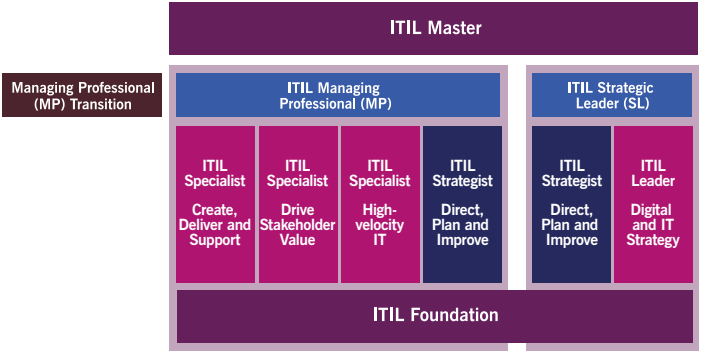


Figure 7.1 The ITIL certification scheme

- IT professionals or others working within an organization that has adopted ITIL.

7.2 ITIL Managing Professional stream

ITIL Managing Professional (ITIL MP) is a stream of four modules that provides practical and technical knowledge about how to run successful IT-enabled services, teams, and workflows.

All modules have ITIL 4 Foundation as a prerequisite.

The target audience consists of:

- IT practitioners working within technology
- digital teams across businesses.

To obtain the designation ITIL Managing Professional, the candidate must complete all four modules in this stream, with ITIL Strategist – Direct, Plan and Improve being a universal module for both streams.

The four modules are:

- ITIL Specialist – Create, Deliver and Support
- ITIL Specialist – Drive Stakeholder Value
- ITIL Specialist – High-velocity IT
- ITIL Strategist – Direct, Plan and Improve.

7.2.1 ITIL Specialist – Create, Deliver and Support

7.2.1.1 Coverage

This module covers:

- core service management activities
- creation of services

- integration of different value streams and activities to create, deliver, and support IT-enabled products and services
- supporting practices, methods, and tools
- service performance
- understanding of service quality and improvement methods.

7.2.1.2 Target audience

Practitioners responsible for:

- managing the operation of IT-enabled and digital products and services
- the end-to-end delivery of services.

7.2.1.3 Training

Accredited training for this module is mandatory.

7.2.2 ITIL Specialist – Drive Stakeholder Value

7.2.2.1 Coverage

This module covers:

- all types of engagement and interaction between a service provider and its customers, users, suppliers, and partners
- conversion of demand into value via IT-enabled services
- key topics such as SLA design, multi-supplier management, communication, relationship management, CX and UX design, and customer journey mapping.

7.2.2.2 Target audience

Practitioners responsible for:

- managing and integrating stakeholders
- customer journey and experience
- fostering relationships with partners and suppliers.

7.2.2.3 Training

Accredited training for this module is mandatory.

7.2.3 ITIL Specialist – High-velocity IT

7.2.3.1 Coverage

This module covers:

- the ways in which digital organizations and digital operating models function in high-velocity environments
- operating in a similar way to successful digitally native organizations
- use of working practices such as Agile and Lean, and technical practices and technologies such as the cloud, automation, and automatic testing, to enable rapid delivery of products and services.

7.2.3.2 Target audience

IT managers and practitioners involved in digital services or digital transformation projects, working within or towards high-velocity environments.

7.2.3.3 Training

Accredited training for this module is mandatory.

7.2.4 ITIL Strategist – Direct, Plan and Improve†

7.2.4.1 Coverage

This module covers:

- creating a 'learning and improving' IT organization, with a strong and effective strategic direction
- influence and impact of Agile and Lean ways of working, and how they can be leveraged to an organization's advantage
- practical and strategic methods for planning and delivering continual improvement.

7.2.4.2 Target audience

Managers at all levels involved in shaping direction and strategy or developing a continually improving team.

7.2.4.3 Training

Accredited training for this module is mandatory.

7.3 ITIL Strategic Leader stream

ITIL Strategic Leader (ITIL SL) is a stream of two modules that recognizes the value of ITIL, not just for IT operations, but for all digitally enabled services. Becoming an ITIL Strategic Leader demonstrates that the individual has a clear understanding of how IT influences and directs business strategy.

† This is a universal module that is a key component of both the ITIL Managing Professional and ITIL Strategic Leader streams.

To obtain the designation ITIL Strategic Leader, the candidate must complete both modules in this stream, with ITIL Strategist – Direct, Plan and Improve being a universal module for both streams. Both modules have ITIL 4 Foundation as a prerequisite.

The two modules are:

- ITIL Strategist – Direct, Plan and Improve
- ITIL Leader – Digital and IT Strategy.

7.3.1 ITIL Strategist – Direct, Plan and Improve

As detailed in section 7.2.4.

7.3.2 ITIL Leader – Digital and IT Strategy

7.3.2.1 Coverage

This module covers:

- alignment of digital business strategy with IT strategy
- how disruption from new technologies is impacting organizations in every industry, and how business leaders are responding
- building and implementing effective IT and digital strategy that can tackle digital disruption and drive success.

7.3.2.2 Target audience

The target audience consists of IT and business leaders and aspiring leaders.

7.3.2.3 Training

Accredited training for this module is mandatory. In addition, those undertaking the ITIL Strategic Leader modules would benefit from a minimum of three years of IT managerial experience.

7.4 ITIL Master

The ITIL Master certification verifies a candidate's ability to apply the principles, methods, and techniques from ITIL in the workplace.

To achieve this certification, a candidate must be able to explain and justify how they have personally selected and applied a range of knowledge, principles, methods, and techniques from the ITIL framework and supporting management techniques, to achieve desired business outcomes in one or more practical assignments.

To be eligible for the ITIL Master certification, a candidate must have both:

- achieved the ITIL v3 Expert certificate or ITIL Managing Professional and ITIL Strategic Leader designations
- worked in IT service management for at least five years in leadership, managerial, or higher management advisory levels.

There is no formal training. PeopleCert, AXELOS's licensed examination institute (EI), provides documentation to support and guide candidates who wish to prepare for the ITIL Master qualification.

7.5 ITIL and the T-shaped individual

The focus of the ITIL 4 advanced-level qualifications has shifted somewhat from previous ITIL v3 education. The deep knowledge of the ITIL processes alone has been enhanced to allow IT service management professionals to gain a wider perspective of their role and interactions with colleagues elsewhere. These modern IT service management professionals are said to have a T-shaped skillset, as shown in Figure 7.2.

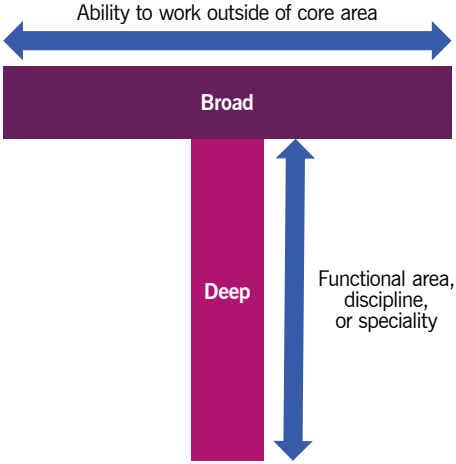


Figure 7.2 A T-shaped skillset

T-shaped skills describe specific attributes of desirable workers. The vertical bar of the 'T' represents expert knowledge and experience in a particular area, such as IT service management. The horizontal bar represents the ability to collaborate with experts in other disciplines, and a willingness to use the knowledge gained from this collaboration.

The T-shaped service manager might look as shown in Figure 7.3.

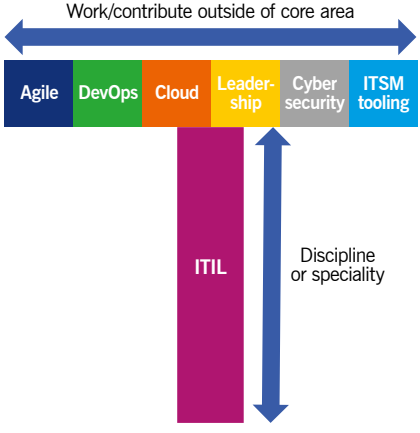


Figure 7.3 A T-shaped service manager

8 ITIL 4: High-velocity IT syllabus

Table 8.1 gives a summary of the concepts that are tested in the examination. It references where these are described, both in the main parts of *ITIL®4: High-velocity IT* and the relevant sections of this guide.

The verb for each assessment criterion indicates the Bloom's level (BL):

- 'Understand' indicates Level 2 understanding/comprehension.
- 'Know how to' indicates Level 3 application.

Table 8.1 Concepts that are tested in the ITIL 4 High-velocity IT examination

Learning outcome		Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
1	Understand concepts regarding the high-velocity nature of the digital enterprise, including the demand it places on IT	1.1	Understand the following terms: a) Digital organization b) High-velocity IT c) Digital transformation d) IT transformation e) Digital product f) Digital technology	a) 2.3 b) 2.1 c) 2.4 d) 2.4.1 e) 2.6.1 f) 2.2	2.2	BL2	4

Table continues

Table 8.1 *continued*

Learning outcome	Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
	1.2	Understand when the transformation to high-velocity IT is desirable and feasible	2.1	2.2.2	BL2	1
	1.3	Understand the five objectives associated with digital products to achieve: a) Valuable investments – strategically innovative and effective application of IT b) Fast development – quick realization and delivery of IT services and IT-related products c) Resilient operations – highly resilient IT services and IT-related products d) Co-created value – effective interactions between service provider and consumer e) Assured conformance – to governance, risk, and compliance (GRC) requirements	a) 2.5.1, 4, 4.1 b) 2.5.1, 4, 4.2 c) 2.5.1, 4, 4.3 d) 2.5.1, 4, 4.4 e) 2.5.1, 4, 4.5	2.3	BL2	2

Learning outcome		Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
2	Understand the digital product lifecycle in terms of the ITIL 'operating model'	2.1	Understand how high-velocity IT relates to: a) The four dimensions of service management b) The ITIL service value system c) The service value chain d) The digital product lifecycle	a) 2.6.6, 2.6.6.1–4, 2.6.7 b) 2.6, 2.6.3, 2.6.3.1, 2.6.3.2, 2.6.4 c) 2.5.1, 2.6.3, 2.6.3.1, 2.6.3.2, 2.6.4 d) 2.6.2	a) 3.1 b) 3.2 c) 3.3 d) 3.4	BL2	3
3	Understand the importance of the ITIL guiding principles and other fundamental concepts for delivering high-velocity IT	3.1	Understand the following principles, models, and concepts: a) Ethics b) Safety culture c) Lean culture d) Toyota Kata e) Lean/Agile/resilient/ continuous f) Service-dominant logic g) Design thinking h) Complexity thinking	a) 3.2.1, 3.2.1.1 b) 3.2.2.2 c) 3.2.3.2, Tab 3.2 d) 3.2.3.3 e) 2.5.2, 2.5.2.1–4, Tab 2.2 f) 2.5.2.5 g) 3.2.1.2 h) 3.2.3.1	a) 4.2.1.1 b) 4.2.2.2 c) 4.2.3.2 d) 4.2.3.3 e) 2.4 f) 2.5 g) 4.2.1.2 h) 4.2.3.1	BL2	3

Table continues

Table 8.1 *continued*

Learning outcome		Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
		3.2	Know how to use the following principles, models and concepts: ● Ethics ● Safety culture ● Lean culture ● Toyota Kata ● Service-dominant logic ● Design thinking ● Complexity thinking To contribute to: a) Help get customers' jobs done b) Trust and be trusted c) Continually raise the bar d) Accept ambiguity and uncertainty e) Commit to continual learning	3.2, 3.2.1, 3.2.1.1, 3.2.2.2, 3.2.3.2, Tab 3.2, 3.2.3.3, 2.5.2, 2.5.2.1–4, Tab 2.2, 2.5.2.5, 3.2.1.2, 3.2.3.1 a) 3.1.4, Tab 3.1 b) 3.1.2, Tab 3.1 c) 3.1.3, Tab 3.1 d) 3.1.1, Tab 3.1 e) 3.1.5, Tab 3.1	4.2.1.1, 4.2.2.2, 4.2.3.2, 4.2.3.3, 2.4, 2.5, 4.2.1.2, 4.2.3.1 a) 4.1.4 b) 4.1.2 c) 4.1.3 d) 4.1.1 e) 4.1.5	BL3	3
4	Know how to contribute to achieving value with digital products	4.1	Know how the service provider ensures valuable investments are achieved	4.1, 4.1.1, 4.1.1.1–3, 4.1.2, 4.1.3, 4.1.4, only high-impact information from Tabs 4.1, 4.2, 4.3, 4.4	2.3.1, 5.1	BL3	1

Learning outcome	Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
	4.2	Know how to use the following practices to contribute to achieving valuable investments: a) Portfolio management b) Relationship management	Tabs 4.1, 4.2, 4.3, 4.4, and the references below which refer to the practice guides: a) 2.1, 2.4 (including subsections) b) 2.1, 2.4 (including subsections)	a) 5.2.1 b) 5.2.2	BL3	2
	4.3	Know how the service provider ensures fast development is achieved	4.2, 4.2.1, 4.2.2, 4.2.3, 4.2.3.1–2, 4.2.4, 4.2.5, 4.2.6, 4.2.7, only high-impact information from Tabs 4.5, 4.6, 4.7, 4.8, 4.9, 4.10, 4.12, 4.13	2.3.2, 5.3	BL3	1

Table continues

Table 8.1 *continued*

Learning outcome	Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
	4.4	Know how to use the following practices to contribute to achieving fast development: a) Architecture management b) Business analysis c) Deployment management d) Service validation and testing e) Software development and management	Tabs 4.5, 4.6, 4.7, 4.8, 4.9, 4.10, 4.12, 4.13 and the references below which refer to the practice guides: a) 2.1, 2.4 (including subsections) b) 2.1, 2.4 (including subsections) c) 2.1, 2.2.2, 2.4 (including subsections), 3.2.2 d) 2.1, 2.1.1, 2.1.2, 2.4 (including subsections) e) 2.1, 2.2, 2.4 (including subsections)	a) 5.4.1 b) 5.4.2 c) 5.4.3 d) 5.4.4 e) 5.4.5	BL3	5

Learning outcome	Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
	4.5	Know how the service provider ensures resilient operations are achieved	4.3, 4.3.1, 4.3.2, 4.3.3, 4.3.4, 4.3.5, 4.3.6, 4.3.7, only high-impact information from Tabs 4.14, 4.15, 4.16, 4.17, 4.18, 4.19, 4.20	2.3.3, 5.5	BL3	1
	4.6	Know how to use the following practices to contribute to achieving resilient operations: a) Availability management b) Capacity and performance management c) Monitoring and event management d) Problem management e) Service continuity management f) Infrastructure and platform management	Tabs 4.14, 4.15, 4.16, 4.17, 4.18, 4.19, 4.20 and the references below which refer to the practice guides: a) 2.1, 2.4 (including subsections) b) 2.1, 2.4 (including subsections) c) 2.1, 2.4 (including subsections) d) 2.1, 2.2.2, 2.4 (including subsections) e) 2.1, 2.4 (including subsections) f) 2.1, 2.4 (including subsections)	a) 5.6.1 b) 5.6.2 c) 5.6.4 d) 5.6.5 e) 5.6.6 f) 5.6.3	BL3	6

Table continues

Table 8.1 *continued*

Learning outcome	Assessment criteria		*ITIL 4: HVIT* and practice guide refs	*HVIT Study Guide* refs	Bloom's level	Marks
	4.7	Know how the service provider ensures co-created value is achieved	4.4, 4.4.1, only high-impact information from Tab 4.21	2.3.4, 5.7	BL3	1
	4.8	Know how to use the following practices to contribute to achieving co-created value with the service consumer: a) Relationship management b) Service design c) Service desk	Tab 4.21 and the practice guides: a) 2.1, 2.4 (including subsections) b) 2.1, 2.4 (including subsections) c) 2.1, 2.4 (including subsections)	a) 5.8.1 b) 5.8.2 c) 5.8.3	BL3	4
	4.9	Know how the service provider ensures assured conformance is achieved	4.5, 4.5.1, 4.5.2, 4.5.3, only high-impact information from Tabs 4.22, 4.23, 4.25	2.3.5, 5.9	BL3	1
	4.10	Know how to use the following practices to contribute to achieving assured conformance: a) Information security management b) Risk management	Tabs 4.22, 4.23, 4.25 and the practice guides: a) 2.1, 2.4 (including subsections) b) 2.1, 2.4 (including subsections)	a) 5.10.1 b) 5.10.2	BL3	2

End notes

1. Derived from Lumiere, in Johnson, P. A. (2014) *Making Light Work*, Fairday Books, Sheffield.

2. Dr Amy Edmondson, Harvard Business School, who coined the term 'psychological safety': www.youtube.com/watch?v=LhoLuui9gX8 [accessed 05 April 2021].

3. Snowden, D. (2011) The Cynefin framework. https://cognitive-edge.com/ [accessed 05 April 2021].

4. England, R.E. (2013) *Plus! The Standard+Case Approach: See Service Response in a New Light*, self-published, ISBN 978-1482061741.

5. Microsoft, Shift left to make testing fast and reliable: https://docs.microsoft.com/en-us/azure/devops/learn/devops-at-microsoft/shift-left-make-testing-fast-reliable [accessed 05 April 2021].

6. Netflix Tech Blog, The Netflix Simian Army: https://medium.com/netflix-techblog/the-netflix-simian-army-16e57fbab116 [accessed 05 April 2021].

7. Abdula, M., Averdunk, I., Barcia, R., Brown, K., and Emuchay, N. (2018) *The Cloud Adoption Playbook: Proven Strategies for Transforming Your Organization with the Cloud*. John Wiley & Sons, Indianopolis.

8. DevOps Audit Defense Toolkit: https://itrevolution.com/devops-audit-defense-toolkit/ [accessed 05 April 2021].

9. Garousi, V. (2010) Applying peer reviews in software engineering education: an experiment and lessons learned. *IEEE Transactions on Education* (Volume 53, Issue 2, May): https://ieeexplore.ieee.org/document/5152941 [accessed 05 April 2021].

Further information

Publications

The ITIL 4 product suite comprises a range of core and supplementary guidance to support you on your IT service management journey. It offers essential interactive digital resources to enhance your learning experience and prepare you for the ITIL 4 High-velocity IT exam, including an official revision app. For more information, visit the AXELOS best-practice website at:

https://www.axelos.com/best-practice-solutions/itil

To buy printed copies or an online subscription to ITIL 4 guidance, visit:

https://www.tsoshop.co.uk/Business-and-Management/AXELOS-Global-Best-Practice/ITIL-4/

Contact points

*it*SMF UK

*it*SMF is the only truly independent and internationally recognized forum for IT service management professionals worldwide. This not-for-profit organization is a prominent player in the ongoing development and promotion of IT service management best practice, standards, and qualifications, and has been since 1991, when the UK chapter commenced as the foundation chapter.

There are more than 50 chapters worldwide. Each chapter is a separate legal entity and is largely autonomous. *it*SMF International provides an overall steering and support function to existing and emerging chapters. It has its own website at www.itsmfi.org.

The UK chapter has more than 5000 members. It offers a flourishing annual conference, online bookstore, annual awards, a professional service management framework (PSMF) competency model, regular member meet-ups, special interest groups, and numerous other benefits. Members of *it*SMF UK can purchase *ITIL®4: High-velocity IT* at tinyurl.com/itsmfitil4

Ground Floor South
Burford House
Leppington
Bracknell RG12 7WW
United Kingdom
Telephone: +44(0) 118 918 6500
Email: membership@itsmf.co.uk
www.itsmf.co.uk

AXELOS

AXELOS is a joint venture company co-owned by the UK Government's Cabinet Office and Capita plc. It is responsible for developing, enhancing, and promoting a number of best-practice methodologies used globally by professionals working primarily in project, programme, and portfolio management, IT service management, and cyber resilience. The methodologies, including ITIL, PRINCE2, PRINCE2 Agile, MSP, RESILIA, and its newest addition, AgileSHIFT, are adopted in more than 150 countries to improve employees' skills, knowledge, and competence in order to make both individuals and organizations work more effectively.

To take you to the next level in your ITIL journey, the ITIL 4 Managing Professional (ITIL MP) suite is available to support the ITIL 4 higher-level and transition exams. For more information, visit:

https://www.axelos.com/certifications/itil-certifications/itil-managing-professional-itil-4

ITIL 4 practice guides are available through My ITIL:
https://www.axelos.com/my-axelos/my-itil

Full details on how to contact AXELOS can be found at https://www.axelos.com.

TSO

TSO (The Stationery Office) is the official publisher for AXELOS, producing official literature for the ITIL portfolio. TSO has a 200-year heritage of secure, no-fail delivery as publisher to the UK Government and Parliament and, more recently, as publisher for many commercial organizations. As part of Williams Lea, a global leader in marketing and communications services, TSO provides key resources for learning and continual development in IT service management.

PO Box 29
Norwich NR3 1GN
United Kingdom

Telephone orders/general enquiries: 0333 202 5070
Textphone: 0333 202 5077
Fax orders: 0333 202 5080
Email: customer.services@tso.co.uk

Index

A

A/B testing 60
Agile techniques 21
AIOps 90
ambiguity 27, 38–39
architecture management practice 75–76
assured conformance
  DevOps Audit Defense Toolkit 107–108
  DevSecOps 108–109
  information security management 110–111
  objective 19–20
  peer review 109–110
  risk management 111–112
  auditors 107–108
  availability management practice 93–94

B

behaviour patterns 3, 38–42
blameless post-mortems 67–68
business analysis 68–69
business analysis practice 76–77
buy/hold/sell technique 58

C

capacity and performance management practice 94–95
causality 50
chaos engineering 86–88
chaos monkey 87–88
ChatOps 91
CI/CD (continuous integration, continuous delivery, and continuous deployment) 33, 69–70
cloud-based services 25–26
co-created value
  objective 19
  relationship management 104
  service design 104–105
  service desk 106–107
  service experience 103
collaborate and promote visibility principle 55
complex adaptive systems 49
complex environments 49–50
complexity thinking 49
containerization 66
continual business analysis 68–69
continual improvement 52–53
continual learning 40
continually raising the bar 39
continuous characteristics 22, 33
continuous delivery 69
continuous deployment 69
continuous integration, continuous delivery, and continuous deployment (CI/CD) 33, 69–70
continuous testing 70–73
cost of delay 57–58
culture *see* organizational culture
customers 40
Cynefin framework 50

D

definition of done 83, 88
deployment management practice 78–79
design thinking 43–44
DevOps 33–34
DevOps Audit Defense Toolkit 107–108
DevSecOps 108–109
digital organizations 10
digital product lifecycles 36–37
digital products 15–16
digital technology 14–15
digital transformation 11–12
digitization 14
done
  definition of 83, 88

E

ethics 42–43
exploration stage of product lifecycle 37
external factors 26–27

F

fast development
  architecture management 75–76
  business analysis 76–77
  CI/CD (continuous integration, continuous delivery, and continuous deployment) 69–70
  continual business analysis 68–69
  continuous testing 70–73
  deployment management 78–79
  infrastructure as code (IaC) 64–65
  Kanban 73–74
  loosely coupled information system architecture 65–66
  objective 17–18
  reviews 66–68
  service validation and testing 80–81
  software development and management 82–85
feedback 33
feedback loops 68–69
focus on value principle 54
four dimensions of service management 24–27
functional outsourcing 26

G

governance and management 30–31

H

'help get customers' jobs done' behaviour 40
High-velocity IT (HVIT)
  approaches 4–5
  behaviour patterns 3, 38–42
  definition 10–11
  examination
    overview 113–114
    question types 114–116
    sample papers 116–117
    syllabus 127–134
    tips 117
  key characteristics 20–23
  objectives 5–9, 16–20
  organizational culture
    behaviour patterns 38–42
    ITIL guiding principles 54–56

models and concepts 3–4
people 41, 44–48
progress 41, 48–53
purpose 41, 42–44
techniques *see* assured conformance; co-created value; fast development; resilient operations; valuable investments
tooling 25
human systems
evolution of 50
HVIT *see* High-velocity IT

I

IaC (infrastructure as code) 64–65
idempotence 65
information and technology dimension of service management 25
information security management practice 110–111
information technology 14
infrastructure and platform management practice 96–97
infrastructure as code (IaC) 64–65
integration of duties 108–109
IT functions 25
IT tooling 25
IT transformation 12–14
ITIL 4
certification scheme 118–126
Foundation certification 118
High-velocity IT
examination
overview 113–114
question types 114–116
sample papers 116–117
syllabus 127–134
tips 117
qualification 113
ITIL continual improvement model 52–53
ITIL guiding principles 4, 54–56
ITIL Managing Professional (ITIL MP) 119–122
ITIL Master 124
ITIL MP (ITIL Managing Professional) 119–122
ITIL practices
architecture management 75–76
availability management 93–94
business analysis 76–77
capacity and performance management 94–95
deployment management 78–79
and HVIT objectives 6–9
information security management 110–111
infrastructure and platform management 96–97
monitoring and event management 98–99
portfolio management 61–62
problem management 100–101
relationship management 63–64, 104
risk management 111–112
service continuity management 101–102
service design 104–105
service desk 106–107

ITIL practices *continued*
service validation and testing 80–81
software development and management 82–85
ITIL service value chain 27, 31–36
ITIL service value system 27–31
ITIL Strategic Leader (ITIL SL) 122–123

K

Kanban 73–74
keep it simple and practical principle 56

L

Lean culture 20–21, 51–52
loosely coupled information system architecture 65–66

M

maintenance 83
management framework 30–31
microservices 66
minimum marketable products 59
minimum viable products and services 58–59
monitoring and event management practice 98–99

O

offboarding stage of product lifecycle 37
onboarding stage of product lifecycle 37
OODA loop 53
operating models 28–29
operational technology 14
optimize and automate principle 56
organizational culture
behaviour patterns 38–42
ITIL guiding principles 54–56
models and concepts 3–4, 40–56
people 41, 44–48
progress 41, 48–53
purpose 41, 42–44
organizations and people dimension of service management 25
outsourcing 26

P

partners and suppliers dimension of service management 25–26
peer review 109–110
performance management 94–95
PESTLE factors 26–27
portfolio management practice 61–62
post-mortems 67–68
prioritization techniques 57–58
problem management practice 100–101
problem-solving 40
processes 26
product lifecycles 36–37
product or service ownership 59–60
progressing iteratively with feedback principle 55
purpose 40–41, 42–44

R

raising the bar 39
reconstructing for service agility 45–46
relationship management practice 63–64, 104
resilient characteristics 22
resilient operations
  AIOps 90
  availability management 93–94
  capacity and performance management 94–95
  chaos engineering 86–88
  ChatOps 91
  definition of done 88
  infrastructure and platform management 96–97
  ITIL practices 93–102
  monitoring and event management 98–99
  objective 18
  problem management 100–101
  service continuity management 101–102
  site reliability engineering (SRE) 92
  technical debt 86
  version control 89
retirement stage of product lifecycle 37
retrospectives 66–67
reviews in fast development 66–68
risk management practice 111–112

S

safety culture 46–47
scrums 83
SDLC (software development lifecycle) model 83
service agility 45–46
service consumers 27, 35–36
service continuity management practice 101–102
service design practice 104–105
service desk practice 106–107
service experience 103
service level agreements (SLAs) 26
service management
  four dimensions of 24–27
service ownership 59–60
service validation and testing practice 80–81
service value chain 27, 31–36
service value system 27–31
service-dominant logic 22–23
Simian army 87
site reliability engineering (SRE) 92
SLAs (service level agreements) 26
software 83
software development 83
software development and management practice 82–85
software development lifecycle (SDLC) model 83
software quality 83
software testing 70–73
SRE (site reliability engineering) 92

start where you are principle 54–55
stress prevention 47–48
suppliers 25–26
systems thinking 49

T

technical debt 83, 86
testing 70–73, 80–81
think and work holistically
  principle 55–56
Toyota Kata 52, 53
trust 39
T-shaped skillsets 124–125

U

uncertainty 27, 38–39

V

valuable investments
  A/B testing 60
  minimum viable products and
    services 58–59
  objective 17
  portfolio management 61–62
  prioritization techniques 57–58
  product or service ownership
    59–60
  relationship management 63–64
value streams 24, 26, 27, 29,
  32–33, 36
value streams and processes
  dimension of service
  management 26
version control 89
VUCA (volatility, uncertainty,
  complexity, and ambiguity) 27

ITIL®4: HVIT Reference and Study Guide will help all those preparing for the Managing Professional ITIL 4 High-velocity IT qualification, and also serve as a quick and easy reference for those who want a high-level authoritative overview of the AXELOS core guidance. With information on the exam and assessment criteria, this is an ideal study tool, authored by the expert who led the HVIT core guidance.

Digital technology is becoming increasingly important. This title focuses on digital products and services, including digital customer experiences, covering effective organizational practices and mental models from a practitioner's perspective.

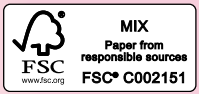
"A great deal of knowledge is distilled in this guide and you will need to unpack it all by reading even more. The knowledge therein has been won by experience and patience, and it has been nicely structured and packaged by Mark, a passionate and seasoned practitioner. Will it work for you? Hopefully it will inspire you to improve, try different things, and open up your conceptual view of the world."
Patrick Debois

This practical guide is perfect for quick and easy learning.

in partnership with
HM Government



HM Government



ISBN 978-0-11-331798-1

